

Mechanical activation of TWIK-related potassium channel by nanoscopic movement and rapid second messenger signaling

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Abstract

Rapid conversion of force into a biological signal enables living cells to respond to mechanical forces in their environment. The force is believed to initially affect the plasma membrane and then alter the behavior of membrane proteins. Phospholipase D2 (PLD2) is a mechanosensitive enzyme that is regulated by a structured membrane-lipid site comprised of cholesterol and saturated ganglioside (GM1). Here we show stretch activation of TWIK-related K⁺ channel (TREK-1) is mechanically evoked by PLD2 and spatial patterning involving ordered GM1 and 4,5-

bisphosphate (PIP₂) clusters. First, mechanical force deforms the ordered lipids, which lowers membrane cholesterol, disrupts the interaction of PLD2 with the GM1 lipids, and allows a complex of TREK-1 and PLD2 to associate with PIP₂ clusters. The association with PIP₂ activates the enzyme, which produces the second messenger phosphatidic acid (PA) that gates the channel. Co-expression of catalytically inactive PLD2 inhibits TREK-1 stretch currents in a biological membrane. Cellular uptake of cholesterol inhibits TREK-1 currents in culture and depletion of cholesterol from astrocytes releases TREK-1 from GM1 lipids in mouse brain. Depletion of the PLD2 ortholog in flies results in hypersensitivity to mechanical force. We conclude PLD2 mechanosensitivity combines with TREK-1 ion permeability to elicit a mechanically evoked response.

Summary

Shear thinning activates TREK-1 through a second messenger.

eLife assessment

This **important** study poses a provocative mechanism of channel activation of the mechanically activated ion channel TREK-1. The data provide **solid** evidence that the application of shear to cells causes a redistribution of both TREK-1 and an associated enzyme, PhospholipaseD2 in the membrane that increases the enzyme activity. The work offers a new mechanism, but note that this is only one possible method of channel activation, and mechanisms independent of PLD2 are also probable.

Introduction

All cells respond to mechanical force, a phenomenon known as mechanosensation (Hahn and Schwartz, 2009 [↗](#); Julius, 2001 [↗](#); Ranade et al., 2015 [↗](#)). Mechanosensation requires the dual processes of sensing and transducing mechanical force into signals that cells can interpret and respond to. Ion channels serve as one type of transducer. While certain channels have been demonstrated to directly respond to mechanical force in purified lipid environments (Brohawn, 2015 [↗](#); Cox et al., 2019 [↗](#); Teng et al., 2015 [↗](#)), others, such as TRP channels, can be downstream components of multi-step signaling cascades (Kwon et al., 2023 [↗](#); Wilde et al., 2022 [↗](#)) in cellular membranes, including those involving mechanosensitive G-protein coupled receptors (GPCRs) (Gudi et al., 1998 [↗](#); Lin et al., 2022 [↗](#); Storch et al., 2012 [↗](#); Wei et al., 2018 [↗](#); Xu et al., 2018 [↗](#)).

One of these channels is TWIK-related K⁺ channel subtype 1 (TREK-1), a mechanosensitive member of the two-pore-domain potassium (K2P) family known for its analgesic properties. TREK-1 plays a role in inhibiting neuronal firing and reducing pain through the release of potassium ions (Honore, 2007 [↗](#)). Intriguingly, the enzyme phospholipase D2 (PLD2) directly interacts with the C-terminus of TREK-1, activating the channel by locally producing phosphatidic acid (PA) (Comoglio et al., 2014 [↗](#)).

Despite lacking a transmembrane domain, PLD2 exhibits mechanosensitivity (Pavel et al., 2020 [↗](#); Petersen et al., 2016 [↗](#)). This property arises from mechanically induced changes in spatial organization of the plasma membrane. Palmitoylations of cysteine residues near its pleckstrin homology (PH) domain, enables PLD2 to bind to a specific site composed of monosialotetrahexosylganglioside (GM1) lipids and cholesterol in the liquid ordered (L_o) region of the membrane (McDermott et al., 2004 [↗](#); Yuan and Hansen, 2023 [↗](#)) (**Supplemental Fig. S1a-b** [↗](#)). Anesthetics compete with palmitates at this site which we call an anesthetic-palmitate (AP) site (Call et al., 2023 [↗](#)). Application of mechanical force disrupts the AP site, releasing the palmitates of PLD2 and allowing the PH domain to interact with phosphatidylinositol 4,5-bisphosphate (PIP₂) clusters in the liquid disordered (L_d) region of the membrane (Petersen et al., 2016 [↗](#)). This, in turn, triggers enzyme activation through spatial redistribution and substrate presentation (Petersen et al., 2020 [↗](#)) (**Supplemental Fig. S1c** [↗](#)).

Notably, the C-terminus responsible for binding PLD2 is essential for TREK-1 mechanosensitivity in cellular membranes (Brohawn, 2015 [↗](#); Chemin et al., 2007a [↗](#); Patel et al., 1998 [↗](#)). Our previous studies showed the same C-terminus is essential for binding PLD2 during anesthetic activation of TREK-1. In that system, the anesthetic-induced release of PLD2 from GM1 lipids was responsible for all the TREK-1 current evoked by anesthetic in HEK293T cells (Pavel et al., 2020 [↗](#)). These results prompted us to investigate whether PLD2's mechanical activation in HEK293T cells contributes to a mechanically evoked TREK-1 current. Analogous to our study on TREK-1 anesthetic sensitivity, here we show that the catalytic activity of PLD2 is indispensable for the generation of mechanically evoked TREK-1 currents in cultured HEK293T cellular membranes. We

further show these PLD2-dependent TREK-1 currents result from the spatial patterning of the channel and enzyme within lipid nanodomains in HEK, N2a, C2C12, and primary neurons. In neurons, this regulatory process is influenced by astrocyte-derived cholesterol.

Results

Dependence on PLD2 for TREK-1 activation

To assess the involvement of PLD2 mechanosensitivity to a mechanically evoked current from TREK-1 channels, we conducted pressure current measurements in HEK293T cells both with and without the expression of a catalytically inactive K758R PLD2 mutant (xPLD2) (Toschi et al., 2009). We selected HEK293T cells for this study due to their minimal endogenous potassium currents, which allowed us to attribute the recorded currents specifically to TREK-1. The expression of TREK-1 was tracked using an EGFP tag attached to the channels C-terminus, and we confirmed successful expression of all constructs, observing their presence on the cell surface (Fig. S2A-B).

In intact HEK293T cells, the overexpression of xPLD2 substantially inhibited nearly all mechanically evoked TREK-1 currents (Fig. 1A-C). When currents were measured in the inside out configuration under negative pressure conditions (from 0 to -60 mmHg), they exhibited a reduction of over 80% from 0.181 ± 0.04 to 0.035 ± 0.013 pA/ μm^2 , despite an overall increase in the total levels of TREK-1 (Fig. S2B-C). This observation suggests that if the effect is through a specific PLD2-TREK-1 interaction, then removal of the PLD2 binding domain should also inhibit PLD2-evoked currents.

To investigate further, we eliminated the putative PLD2 binding site on TREK-1 by truncating the C-terminus at residue 322 (TREK trunc) (see Fig. S3A). The pressure induced current in TREK trunc decreased by more than 85%, measuring 0.025 ± 0.013 pA/ μm^2 (Fig. 1C and Fig. S3B). Remarkably, this reduction was nearly identical to the results obtained when full-length TREK-1 (TREK FL) was expressed in conjunction with xPLD2. Importantly, previous studies have confirmed the functionality and mechanosensitivity of this similar truncated TREK-1 when reconstituted into crude soy PC lipids (Stephen G Brohawn et al., 2014).

Conversely, the overexpression of TREK-1 with wild-type (WT) mouse PLD2 (mPLD2) resulted in a substantial augmentation of TREK-1 pressure-dependent currents within HEK293T cells (Fig. 1A, see Supplemental Fig. S3B for raw traces). The TREK-1-evoked currents were markedly significant, both with and without overexpression of mPLD2, when compared to xPLD2 ($p < 0.002$ and 0.007 respectively; Fig. 1C). HEK293T cells, like all cells, have endogenous PLD2 (enPLD2). The xPLD2 expression is >10x the endogenous PLD2 (enPLD2) which we previously showed outcompetes the binding of enPLD2 to TREK-1 under similar conditions (Pavel et al., 2020).

The threshold for PLD2-dependent TREK-1 activation proved to be quite sensitive. TREK-1 exhibited responsiveness to negative pressure (ranging from 0 to -60 mmHg, in a non-saturating regime) with a half maximal pressure requirement of approximately 32 mmHg. This pressure led to the generation of up to 200 pA of TREK-1 current (Fig. 1A and Supplemental Fig. S3C), aligning with findings from prior investigations (Stephen G Brohawn et al., 2014; Patel et al., 1998).

As previously mentioned, previous studies demonstrated TREK-1 direct sensitivity to mechanical force in purified lipids, independent of PLD2 (Berrier et al., 2013; Stephen G Brohawn et al., 2014). To delineate a PLD2-independent (i.e., direct) component of TREK-1 mechanotransduction within a cellular membrane we conducted a comparative analysis between cells expressing TREK trunc and mock transfected HEK293T cells (lacking TREK-1 expression). Our

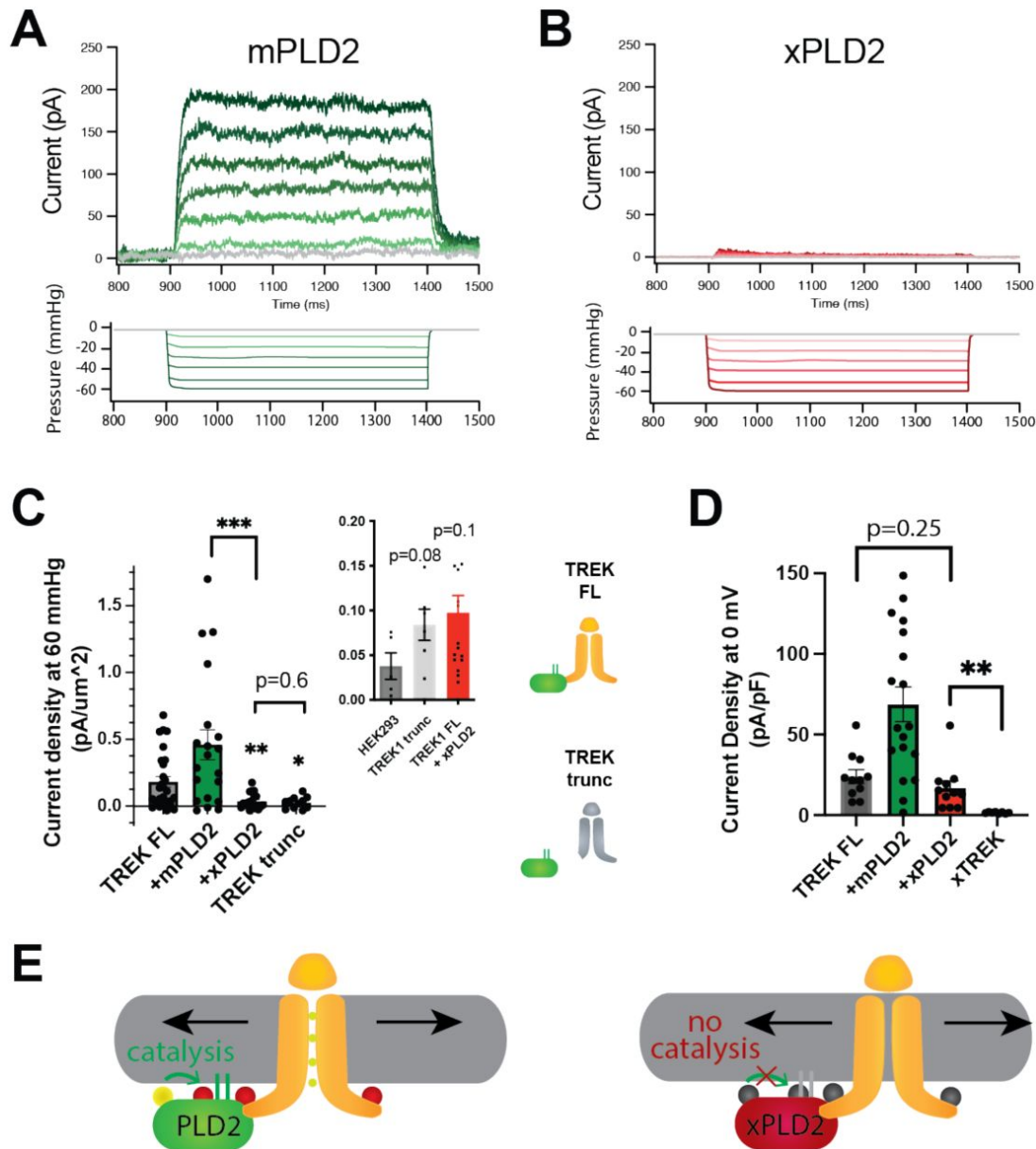


Figure 1.

PLD2-dependent and independent mechanical activation of TREK-1 channels.

(A-B) Representative traces from pulled patches of human TREK-1 over expressed in HEK293T cells with mouse phospholipase D2 (mPLD2, green traces) (A) or catalytically inactive mouse PLD2 (xPLD2, red) (B) under pressure clamp (0-60 mmHg at +30 mV). (C) The data, after subtracting HEK293T background current (0.04 ± 0.02 pA/ μm^2 $n=5$ (inset)), are summarized for -60 mmHg. Compared to endogenous PLD2, the expression of xPLD2 eliminated the majority of detectable TREK-1 pressure current ($p<0.007$, $n=16-23$), as did a functional truncated TREK-1 (TREK trunc) lacking the PLD2 binding site ($p=0.002$, $n=15-23$). The inset compares mock-transfected HEK293T cells with TREK trunc and full length TREK-1 (TREK FL) + xPLD2, indicative of direct TREK-1 activation. Asters indicate significance relative to TREK FL, except where noted by a bar. (D) Whole cell TREK-1 potassium currents with and without xPLD2. TREK-1 is expressed and functional in the presence of xPLD2. A nonfunctional C-terminal truncation (C321) of TREK-1 (xTREK) is shown with no appreciable current HEK293T cells. (E) Cartoon illustrating PLD2-dependent TREK-1 opening in HEK293T cellular membrane. On the left, membrane stretch (black arrows) mechanically activates PLD2. When PLD2 is active, it makes phosphatidic acid (PA) which evokes the open state of TREK-1. On the right, in the absence of mechanically generated PA, the closed channels remain closed despite the presence of membrane tension. Statistical comparisons were made with an unpaired Student's *t* test.

results demonstrated that TREK trunc exhibited a modest increase in current, measuring $0.046 \pm 0.023 \text{ pA}/\mu\text{m}^2$ ($p=0.08$, **Fig. 1c** [inset](#)), when compared to the condition with no TREK-1. Similarly, in cells expressing xPLD2 alongside full-length TREK-1, the channel exhibited a small increase in current measuring $0.060 \pm 0.035 \text{ pA}/\mu\text{m}^2$ ($p=0.11$). This PLD2 independent current aligns with direct mechanical activation of TREK-1 observed in liposomes. However, it is important to note that we cannot rule out the possibility of other contributing mechanisms.

Due to the minimal pressure-activated TREK-1 current observed in the presence of xPLD2, we conducted control experiments to validate the presence of functional TREK-1 on the plasma membrane—channels capable of conducting current. TREK-1 is a potassium leak channel with inherent basal currents in cultured cells ([Comoglio et al., 2014](#) [inset](#)). It is reasonable to assume that the basal current arises from the presence of basal levels of anionic lipids in the cellular membrane.

As anticipated, we observed TREK-1 basal current both in the presence ($16.9 \pm 4.3 \text{ pA/pF}$) and absences ($24.0 \pm 4.2 \text{ pA/pF}$) of xPLD2 ([Comoglio et al., 2014](#) [inset](#)) (**Fig. 1d** [inset](#)). These basal resting currents were recorded at 0 mV in HEK293T cells over expressing TREK-1 FL with and without xPLD2. Notably, the current in the presence of xPLD2 was significantly higher than that observed in a nonfunctional control TREK-1 (xTREK), which we previously determined to exhibit no measurable current in unrelated studies. These controls affirm that TREK-1 FL is expressed and functionally active in the presence of xPLD2. Moreover, they highlight the essential role of PLD2 in pressure-activated currents (**Fig. 1c** [inset](#)), contrasting with the basal leak currents (**Fig. 1d** [inset](#); TREK FL, grey bar vs. +xPLD2, red bar), which did not necessitate PLD2.

Mechanical activation of TREK-1 by movement between nanodomains

As previously reported in a prior publication, we proposed that fluid shear induces the disassociation of PLD2 with cholesterol dependent GM1 clusters and the association with PIP₂ clusters, leading to a change in spatial organization and activation of the enzyme (**Fig. S1** [inset](#)) ([Petersen et al., 2016](#) [inset](#)). Given that TREK-1 binds to PLD2, we hypothesized the channel could undergo a similar mechanically induced spatial reorganization between GM1 and PIP₂ clusters. To visualize this spatial patterning of TREK-1 in response to shear forces, we developed a method to chemically fix membranes during shear (**Fig. 2A** [inset](#)). In our experiments, we employed cultured HEK293T cells expressing both TREK-1 and mPLD, which matched the conditions used in our electrophysiology experiments. These cells were subjected to shear forces of 3 dynes/cm^2 , a physiologically relevant force to cells ([Petersen et al., 2016](#) [inset](#); [Schneck, 1992](#) [inset](#)), using a rotary shear approach. Subsequently, the cells were fixed and labeled for two- or three-color dSTORM. TREK-1 proteins were tagged with EGFP, leveraging its inherent self-blinking properties for detection ([Call et al., 2023](#) [inset](#)). Lipids were labeled using Alexa 647 (A647) anti-PIP₂ antibodies or A647-labeled cholera toxin B (CTxB), which selectively bind to PIP₂ and GM1 clusters, respectively ([Petersen et al., 2016](#) [inset](#)). By concurrently labeling both lipids and proteins, we were able to monitor the dynamic changes in spatial organization within nanoscopic lipid domains ([Yuan and Hansen, 2023](#) [inset](#)). To access the PIP₂ domains on the inner leaflet, cells were permeabilized (see methods). The effectiveness of cellular staining was confirmed through confocal microscopy (see **Fig. S2A-G** [inset](#)).

Before the application of shear forces, TREK-1 and GM1 lipids displayed a robust correlation (**Fig. 2B, C** [inset](#)), suggesting a close association between TREK-1 and GM1 lipids. However, after the application of shear forces, this correlation significantly diminished ($p<0.01$ at 50 nm). A similar experiment using a cy3b-labeled anti-TREK-1 antibody yielded nearly identical results, confirming the validity of both our EGFP-dSTORM method and the specificity of the TREK-1 antibody (**Fig. S4A** [inset](#)) (**Fig. S2A** [inset](#)).

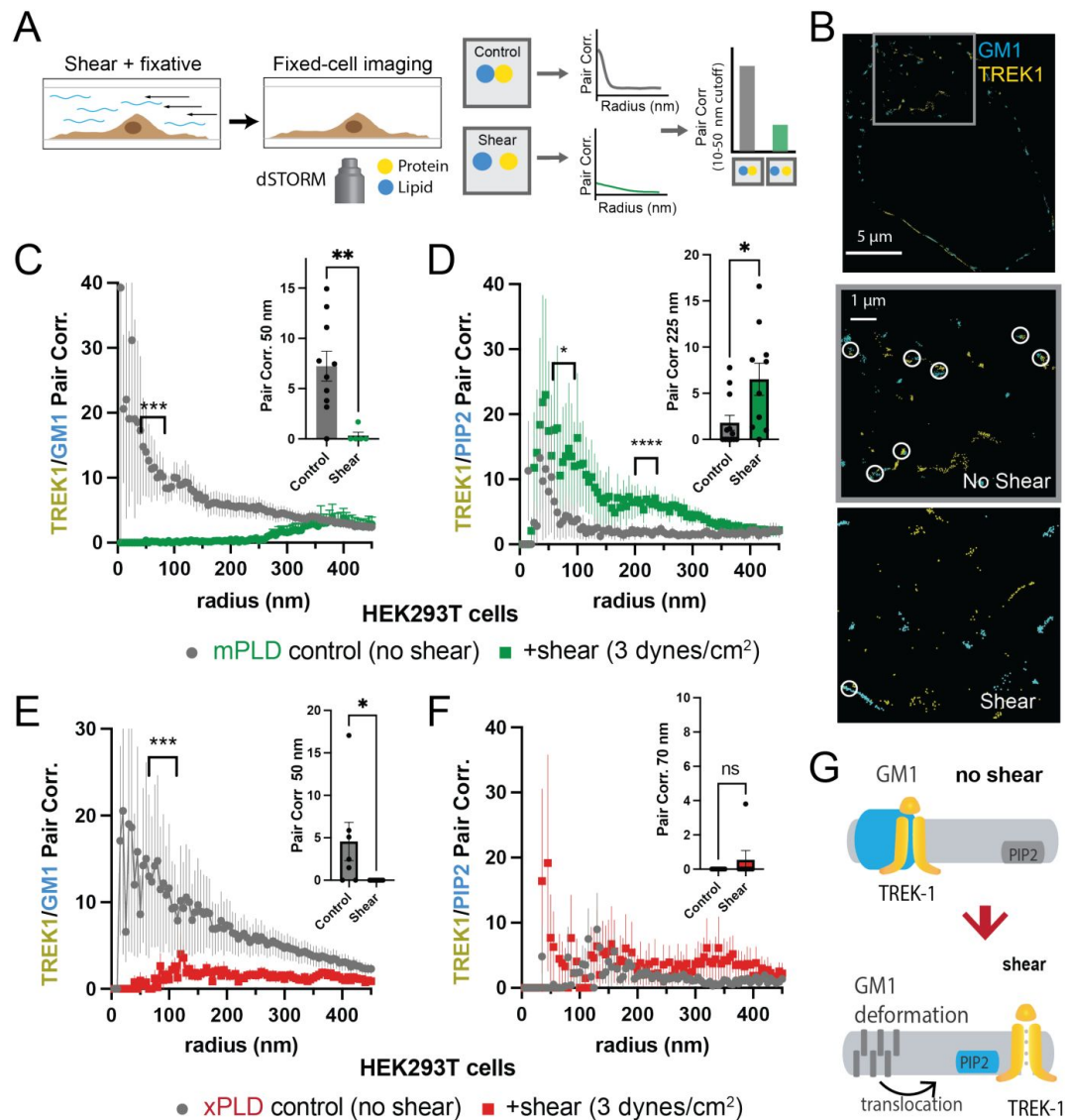


Figure 2

Shear induces nanoscopic movement of TREK-1 in HEK293T cells.

(A) Schematic representation of the shear fixing protocol. Cells grown in a shear chamber are fixed while shear force is applied. Fixed samples are then labeled with fluorescent antibodies or CTxB and subjected to imaging for nanoscopic movement (<250 nm) by two-color super resolution imaging and pair correlation (Pair corr.). **(B)** EGFP-STORM imaging of TREK-1: EGFP and Alexa 647 cholera toxin B (CTxB) with and without shear in HEK293T cells. The middle panel, outlined in grey, is a zoomed portion of the cell surface outlined in the top panel. The bottom panel is a zoomed portion of the cell surface from a cell treated with shear (See Fig. S2I for full image). Locations of TREK-1/GM1 proximity are outlined with a white circle. **(C)** Pair correlation analysis (Pair corr.) of TREK-1 with GM1 lipids before and after shear (3 dynes/cm²; green) determined by EGFP-STORM imaging when mouse phospholipase D2 (mPLD2) is over expressed (non-permeabilized). The significance of the Pair corr. change is shown across the range of radii 50-70 nm (along the curve) and at a single 50 nm radius (inset). **(D)** Combined EGFP-STORM imaging of TREK-1 with Alexa 647 labeled PIP₂ in the presence of over expressed mPLD2 (permeabilized). Significance is shown for radii 70-85 nm along the curve and at a single 225 nm radius (inset). **(E-F)** Combined EGFP-STORM of TREK-1 in the presence of catalytically inactive PLD2 (xPLD2). Shear (3 dynes/cm²) of TREK-1 is shown as a red curve with xPLD2 present. The experiments are as described in panels c-d. In **(E)** a significant shift in TREK-1/GM1 Pair corr. is shown for 50-70 nm (along the curve) and at a 50 nm radius (inset). In **(F)** Pair corr. did not appear to shift significantly, as determined by a Student's t-test or for multiple point a nested Student's t-test; *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. **(G)** Cartoon illustrating the association of TREK-1 with GM1 lipids prior to shear (top) and with PIP₂ lipids (bottom) in response to mechanical shear (red arrow).

Following the release of TREK-1 from GM1 lipids, we anticipated that it would undergo a nanoscale repositioning toward PIP₂ clusters. PIP₂ forms distinct nanodomains that are separate from GM1 clusters (Petersen et al., 2016 [↗](#); van den Bogaart et al., 2011 [↗](#); Wang and Richards, 2012 [↗](#)) by an average distance of approximately 133 nm in HEK293T cells (Yuan et al., 2022 [↗](#)). Cells were subjected to shear forces of 3 dynes/cm², permeabilized, fixed, and subsequently labeled with an anti-PIP₂ antibody. As predicted, the correlation between TREK-1 and PIP₂ was initially low prior to shear application, but it significantly increased after shear forces were applied (**Fig. 2D** [↗](#)), in contrast to the observations made with GM1 (**Fig. 2C** [↗](#)). Therefore, shear forces induce TREK-1 to disassociate from GM1 clusters and associate with PIP₂ clusters. Statistical analysis using a Student's t test confirmed the significance of TREK-1 clustering with PIP₂ at both a single radius and multiple point comparisons along the curve.

Subsequently, we investigated the rearrangement of TREK-1 induced by shear forces in the presence of xPLD2. Like mPLD2, we overexpressed xPLD2 in HEK293T cells, mirroring the conditions employed in our electrophysiology experiments in **Fig. 1B** [↗](#). Furthermore, as with mPLD2, HEK293T cells were subjected to shear forces, permeabilized, labeled, and imaged with two-color dSTORM.

In the presence of xPLD2, shear caused TREK-1 to leave GM1 domains (**Fig. 2E** [↗](#)). However, unlike the response observed with mPLD2, the association of TREK-1 with PIP₂ clusters remained relatively weak following shear forces (**Fig. 2F** [↗](#)), despite an overall increase in TREK-1 and PIP₂ levels in the membrane prior to shear (**Fig. S2C-D** [↗](#)). **Figure 2G** [↗](#) shows a model of shear induced movement of TREK-1 from GM1 to PIP₂ clusters.

Mechanism of PLD2 activation by shear

It is presumed that TREK-1 translocates between nanodomains as a complex with PLD2 (Comoglio et al., 2014 [↗](#)). Lacking its own palmitoylation, TREK-1 interacts with PLD2 through its unstructured C-terminus. In earlier research, we demonstrated that cholesterol drives PLD2 associates with GM1 lipids. The application of mechanical force activated PLD2, presumably through the release of PLD2 from GM1 lipids. However, this PLD2 patterning under mechanical shear has not been directly demonstrated with dSTORM.

To validate our proposed mechanism regarding the shear-induced movement of PLD2 with dSTORM, we employed calibrated shear chambers (ibidi μ -Slide I^{0.4} parallel-plate) with cultured C2C12 muscle cells (mouse myocytes) and N2a mouse neuroblastoma cells, which naturally express TREK-1 (see **Fig. S2E** [↗](#)). Utilizing endogenous expression helped circumvent potential artifacts that may arise from artificially saturating GM1 clusters through protein overexpression.

Our shear experiments were initiated by perfusing phosphate buffered saline (PBS) through the calibrated shear chambers, applying a precisely controlled force of 3 dynes/cm². To maintain a constant temperature of 37 °C, we employed a digitally controlled inline heater. Immediately following the application of shear forces (within <10 seconds), we introduced fixative agents into the shear buffer, facilitating the rapid fixation of cells in their mechanically stimulated state. This approach paralleled the methodology used in our experiments involving rotary shear (see **Fig. 3a** [↗](#), **S5a** [↗](#)), but with tighter control of temperature.

Utilizing two-color dSTORM and analysis by the pair correlation function, we observed that shear forces introduce the mobilization of PLD2 within the cell membrane, and this effect was independent of temperature fluctuations. Prior to the application of shear forces, PLD2 exhibited a strong association with GM1 clusters (**Fig. 3B** [↗](#)). However, after the application of shear forces, the correlation of PLD2 with GM1 clusters decreased significantly, while it robustly increased in association with PIP₂ clusters (**Fig. 3C** [↗](#)). Notably, the release of PLD2 from GM1 domains induced by shear forces was more pronounced than the disruption caused by anesthetic agents, as

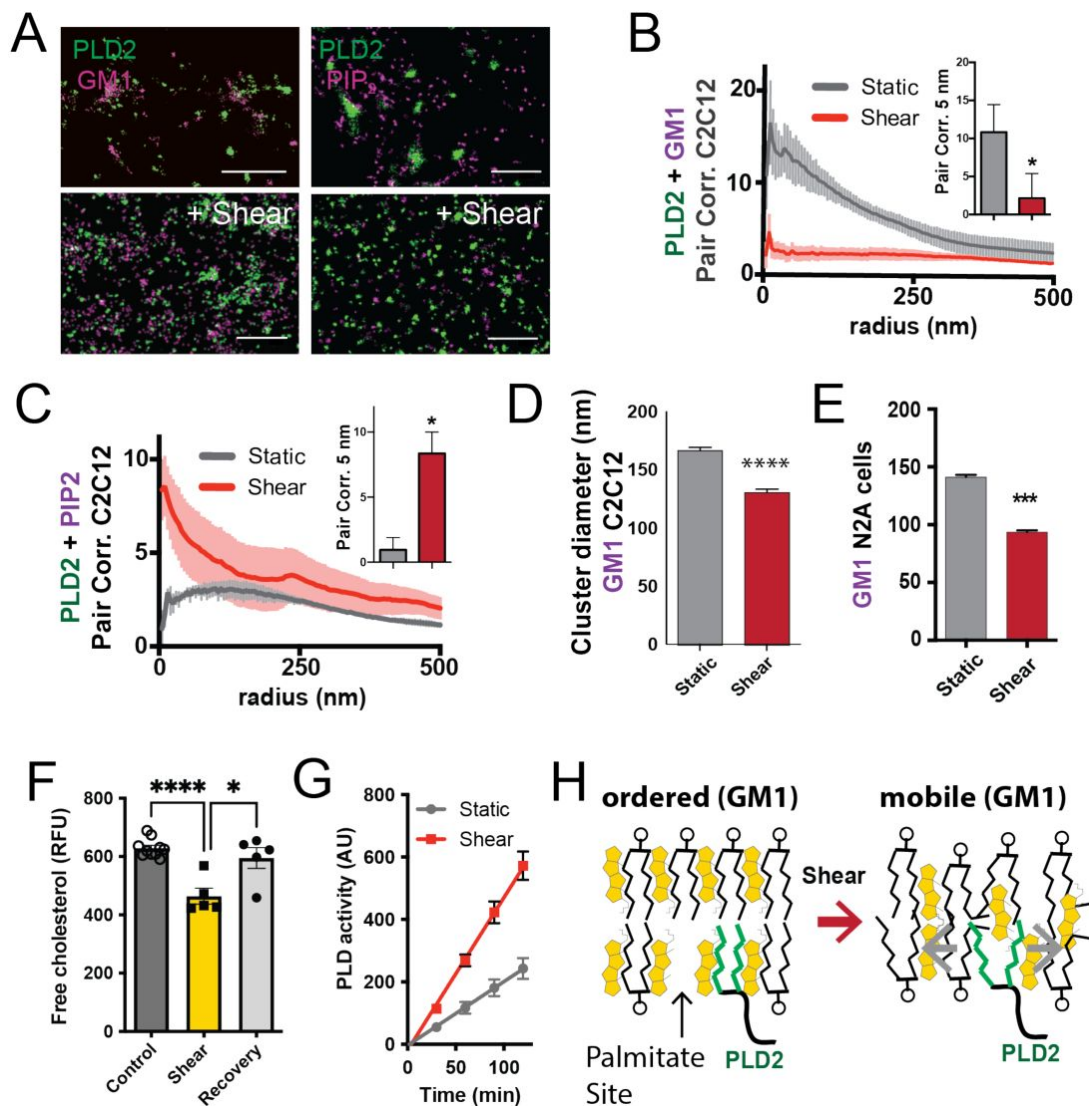


Figure 3.

Shear mobilizes PLD2 within ordered GM1 lipids.

(A) two-color dSTORM images of fixed C2C12 cells with and without (3 dynes/cm²) shear. Cells were labeled with fluorescent CTxB (ganglioside GM1) or antibodies (anti-PIP₂ or anti-PLD2 as indicated) and sheared with the temperature held constant at 37 °C. Scale bar = 1 μ m. (B) Pair correlation analysis (Pair corr., unitless) of PLD2 with GM1 or PIP₂ lipids at a given radius. Error bars are displayed at a given radius. A bar graph (inset) is shown at the shortest calculated radius of 5 nm (single point on the x axis). Prior to shear (grey line), PLD2 associates with GM1 clusters; after shear, there is almost no association. (C) The opposite was true for phosphatidylinositol 4,5 bispophosphate (PIP₂). Prior to shear, PLD2 does not associated significantly with PIP₂ clusters, after shear, its association increases dramatically. (D) Cluster analysis of the GM1 lipids from the C2C12 cells shown in (a). (E) Cluster analysis of GM1 lipids in neuroblastoma 2a (N2a) after 3 dynes/cm² shear force. (F) Fluorescent cholesterol assay. N2a cells grown in a 48 well plates were sheared with 3 dynes/cm² orbital fluid shear, fixed with shear (10 min), and compared to control cells with no shear using a fluorescent cholesterol assay. After shear, a second set of control cells were allowed to recover with no shear and fixative for 30s (recovery), otherwise the cells were treated identical to experimental cells. (n=5-10). (G) A live PLD activity assay demonstrates that fluid shear (3 dynes/cm²) increases substrate hydrolysis in cultured N2a cells. (n>800 clusters from 5-6 cells). (H) Depiction of shear thinning activating PLD2. (left) The palmitates of PLD2 (green lines) are shown bound to the palmitate site in ordered GM1 lipids. (right) After shear cholesterol is reduced, and GM1 lipids are deformed. The deformed surface no longer binds palmitates efficiently allowing the palmitates move freely—a process known as shear thinning. Statistical comparisons were made with an unpaired Student's t-test (*p<0.05, ***p<0.001, ****p<0.0001).

previously reported (Pavel et al., 2020 [DOI](#)). The stability in temperature, maintained within a range of $\pm 0.1^\circ\text{C}$, suggests the mechanism is unlikely to be attributed to the thermal melting of ordered lipids near a phase transition state.

Conducting cluster analysis of GM1 particles subjected to shear revealed a notable reduction in the apparent size of GM1 clusters, decreasing from 167 ± 3 to 131 ± 3 nm (approximately 20%, **Fig. 3d** [DOI](#)) in C2C12 cells. A similar effect on the size of GM1 clusters was also observed in N2a cells (**Fig. 3E** [DOI](#)). PIP₂ clusters also remained largely intact with a reduction in size (from 154 ± 1 to 139 ± 1) after shear (**Fig. 55B** [DOI](#)).

The observed reduction in cluster size closely resembled the outcomes of prior experiments in which we depleted cholesterol with methyl-beta-cyclodextrin (MBCD) (Pavel et al., 2020 [DOI](#); Petersen et al., 2016 [DOI](#)). Given that MBCD is a known cholesterol removal agent, we conducted assays on N2a cells subjected to orbital fluid shear (3 dynes/cm²) to examine changes in cholesterol levels. The cells were subjected to 10 min of shear at 37 °C, followed by fixation.

Our results revealed a 25% reduction in free cholesterol levels in N2a cells exposed to orbital shear ($p < 0.001$) (**Fig. 3F** [DOI](#)), which coincided with the activation of PLD (**Fig. 3G** [DOI](#)). Importantly, this decrease in cholesterol was statistically significant $p < 0.0001$ and reversible. Allowing the cells to recover for approximately 30 seconds prior to fixation resulted in the restoration of cholesterol levels to those observed in non-sheared cells (**Fig. 3F** [DOI](#)).

In C2C12 cells, TREK-1 and PLD2 exhibited a strong correlation both before (**Fig. 55C** [DOI](#), grey trace) and after shear (**Fig. 55C** [DOI](#), red trace), providing further evidence that they form a complex at least intermittently, in both shear and unsheared states. Following shear, a minor fraction of TREK-1 was found to associate with PIP₂ (**Fig. 55D** [DOI](#)). Interestingly, in contrast to HEK293T cells, C2C12 cells displayed minimal TREK-1/GM1 correlation (**Fig. 55E** [DOI](#), grey trace). Notably, PLD2 regulation is known to be influenced by cholesterol (Petersen et al., 2016 [DOI](#)). In various cell types, we have observed that cultured cells tend to have lower cholesterol levels than human tissues (Wang et al., 2023 [DOI](#), 2021 [DOI](#); Yuan et al., 2022 [DOI](#)). These observations lead us to consider the possibility of higher levels of cholesterol in our experimental setup for cultured cells.

Regulation of TREK-1 clustering by cholesterol and GM1 lipids

Cholesterol levels, especially in the brain, can be notably high (Hansen, 2023 [DOI](#); Zhang and Liu, 2015 [DOI](#)). We hypothesized that cholesterol might influence the endogenous TREK-1 to associate with PLD2 in GM1 clusters. To test this hypothesis, we introduced cholesterol into C2C12 cells using apolipoprotein E (apoE) lipidated with 10% serum, as apoE is a naturally occurring cholesterol transport protein (Wang et al., 2023 [DOI](#), 2021 [DOI](#); Yuan et al., 2022 [DOI](#)) (see **Fig. 4A** [DOI](#)). Remarkably, lipidated apoE induced a substantial clustering of TREK-1 with GM1 lipids in membranes of C2C12 cells (**Fig. 4B** [DOI](#)). This effect was also observed with TREK-1 in N2a cells. Intriguingly, the application of fluid shear (3 dynes/cm²) completely reversed the effect of cholesterol (**Fig. 4C** [DOI](#)).

We anticipated that the cholesterol-induced association of TREK-1 with GM1 lipids would lead to a reduction in TREK-1 currents, as PLD2 would be inhibited due to a lack of substrate. To directly assess the activity of TREK-1 under conditions of elevated cellular cholesterol, we overexpressed TREK-1 in HEK293T cells and quantified the current density in whole cell patch clamp mode, both with and without cholesterol uptake (**Fig. 4D** [DOI](#)). Consistent with our proposed model, TREK-1 current density decreased nearly 2.5-fold in cholesterol-loaded cells, and this reduction in current was statistically significant ($p < 0.05$).

In our earlier investigation, we established that astrocytes play a pivotal role in regulating the clustering of proteins in neurons. This regulation is achieved through the release of apoE-containing particles, which are primarily laden with cholesterol. Remarkably, our studies revealed

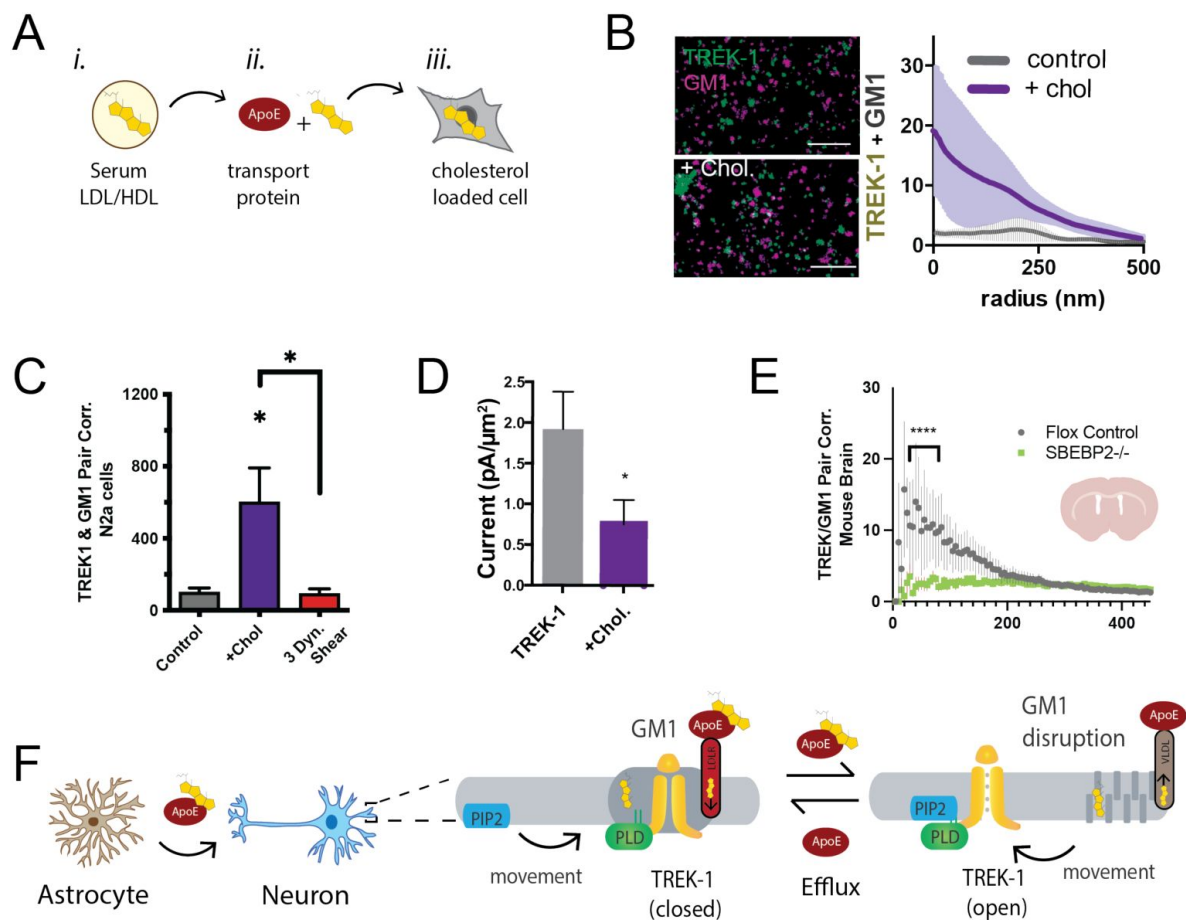


Figure 4.

Astrocyte cholesterol regulates TREK-1 through GM1 lipids and spatial patterning.

(A) Uptake of cholesterol into cultured cells using the cholesterol transport protein apolipoprotein E (apoE). (B) Cholesterol/lipid uptake into C2C12 cells with 4 $\mu\text{g}/\text{ml}$ (~ 110 nM apoE, purple line). Cholesterol dramatically increases TREK-1 correlation of TREK-1 with GM1 labeled lipids. Without cholesterol (grey line) very little TREK-1 clusters with GM1 lipids. Scale bars = 1 μm . (C) Pair correlation (Pair corr.) of TREK-1 and CTxB, localized within 5 nm of each other, are shown plotted after cholesterol treatment (apoE + serum, purple shading) or treatment with 3 dynes/ cm^2 shear (red shading). Cholesterol increased TREK-1 association almost 5-fold and shear reversed the effect. ($n=4-7$); unpaired Student's t test (D) Current densities from whole cell patch clamp recordings are shown with and without cholesterol loading with 4 $\mu\text{g}/\text{ml}$ apoE in HEK293T cells over expressing human TREK-1. Increasing cholesterol inhibited the channel ~ 3 -fold (Student's t-test; $p<0.05$) (E) Reduction in neuronal cholesterol results in a decrease in correlation between TREK-1 and GM1 cluster in brain slices from a hGFAP-Cre driving the SB2 knockout mouse and its Flox control. Student's t-test at 25nm $*p<0.05$, nested Student's t-test at 25-50 nm ($****p<0.0001$, $n=20-24$ unspecified cortical regions). (F) Proposed model for cluster associated TREK-1 activation and inhibition. In high cholesterol, TREK-1 clusters with PLD2 and ordered (thick) GM1 lipids inhibiting the channel. In low cholesterol, TREK-1 is partially clustered closer to PIP₂ generating basal TREK-1 activity (see also Fig. 1C). After shear, the order of GM1 clusters (dark grey) is disrupted further increasing PLD2 and TREK-1 clustering with PIP₂ lipids (blue).

that cholesterol amplifies the clustering of proteins within GM1 domains. Furthermore, we demonstrated that disrupting the cholesterol synthesis in astrocytes leads to the reversal of protein clustering, as previously reported (Wang et al., 2021 [DOI](#)). It stands to reason that the same astrocytic cholesterol dynamics could govern the localization of TREK-1 within GM1 lipids in the brain of a mouse.

To investigate the *in vivo* regulation of TREK-1 clustering by astrocytes, we stained 50-micron brain slices obtained from both control mice and mice specifically engineered to lack cholesterol synthesis in astrocytes. The depletion of cholesterol from astrocytes was achieved by targeting sterol regulatory element-binding protein 2 (SREBP2), a crucial regulator of cholesterol synthesis (Ferris et al., 2017 [DOI](#); Wang et al., 2021 [DOI](#)). Before slicing, the brains underwent fixation via whole-body perfusion. The resulting free-floating brain slices were mounted onto circular cover slips using a fiberglass filter paper, which facilitated dSTORM buffer access to the tissue with minimal background interference (see **Fig. S4E** [DOI](#)). These brain slices, taken from coronal sections near the hippocampus were subjected to staining with cy3b-anti-TREK-1 antibody and A647-CTxB. The imaging process encompassed multiple unspecified locations (approximately 10 regions), and subsequent analysis was performed employing the pair correlation function.

In flox control mice (wild type SREBP2), TREK-1/GM1 correlation exhibited a distinct pattern, with TREK-1 clustering near GM1 (**Fig. 4E** [DOI](#)), aligning with the GM1 correlation seen in N2a cells with elevated levels of cholesterol (**Fig. 4B** [DOI](#)). As anticipated, in the brains of mice with astrocyte-specific SREBP2 deficiency, TREK-1 displayed a reduced association with GM1 lipids ($p < 0.0001$). These findings serve as compelling evidence that, in the brain of an animal, astrocytic cholesterol serves as a critical regulator of TREK-1's affinity for inhibitory GM1 clusters.

Figure 4F [DOI](#) presents a comprehensive model detailing the regulation of TREK-1 by astrocyte-derived cholesterol, using the insights derived from both cultured cells and *ex vivo* studies. In this model, cholesterol originating from astrocytes is transported to neurons via apoE. Under conditions of elevated membrane cholesterol, TREK-1 forms associations with inhibitory GM1 lipids, a scenario in which PLD2 has limited substrate availability. Conversely, when membrane cholesterol levels are reduced, TREK-1 relocates away from GM1 lipids toward activating PIP₂ lipids. In this context, PLD2 gains improved access to its substrate phosphatidylcholine (PC), leading to the production of lipid agonists such as PA, which ultimately evoke activation of the TREK-1 channel.

Direct activation of PLD2 by osmotic stretch

Osmotic stretch is a well-documented activator of TREK-1 (Patel et al., 1998 [DOI](#)). To investigate whether osmotic stretch also activates PLD2, we conducted experiments to monitor PLD product release in live cells subjected to hypotonic stretch. Our findings revealed that membrane stretch by a 70 mOsm swell (representing a high degree of stress) led to approximately 50% increase in PLD2 activity in N2a cells (**Fig. 5A** [DOI](#)). It is worth noting that this level of activation, although significant, was less pronounced compared to shear stress in N2a cells (**Fig. 3E** [DOI](#)), which increased approximately 3-fold in activity.

Subsequently, we conducted dSTORM experiments to investigate whether stretch had the capacity to release PLD2 from GM1 lipids within the membranes of N2a cells. In these experiments, cells were exposed to either a hypotonic solution with an osmolality of 70 mOsm (indicating a state of swelling) or an isotonic solution with an osmolality of 310 mOsm (representing a control condition) for 15 min at 37 °C. Following this treatment, the cells were fixed, labeled, and subjected to imaging using the same procedures employed in the shear stress experiments. Our results, consistent with the outcomes of shear stress experiments and PLD2 assay, clearly demonstrated a discernible shift in the spatial distribution of PLD2, transitioning from ordered GM1 clusters to the PIP₂ clusters after osmotic stretch (**Fig. 5B-D** [DOI](#)).

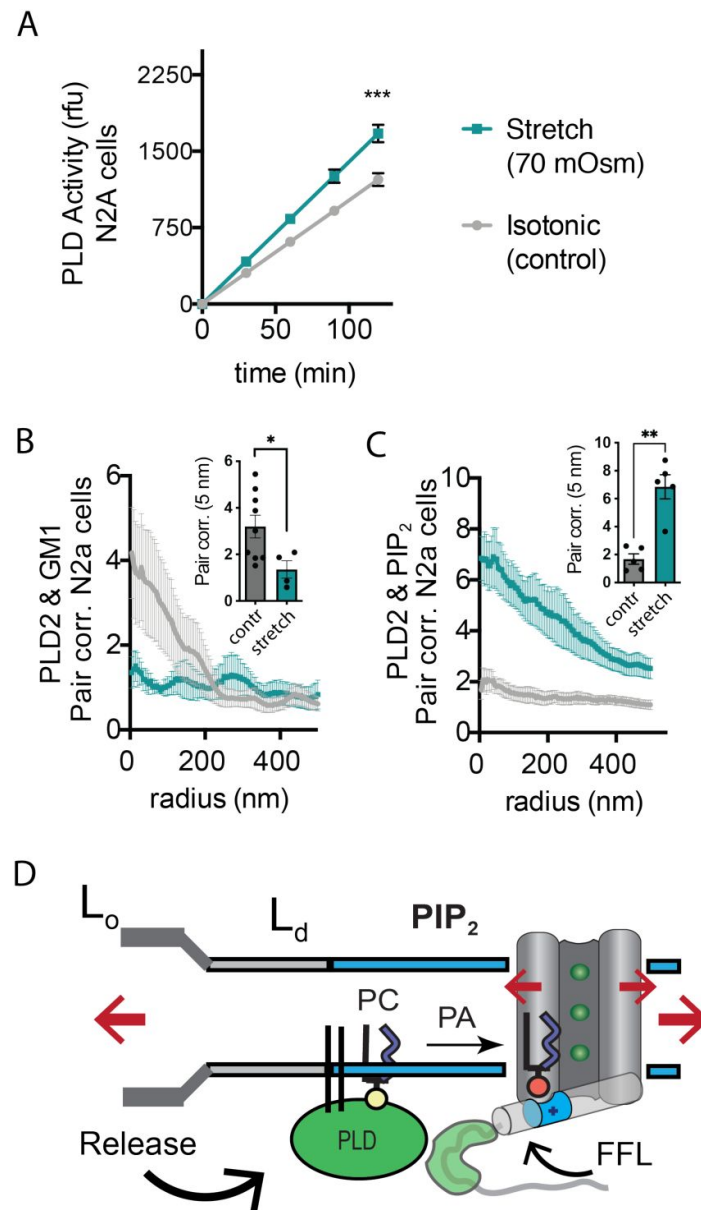


Figure 5.

Osmotic stretch activates PLD2 in N2a cells.

(A) Stretch by osmotic swell (70 mOsm buffer) increased PLD2 catalytic activity in live neuroblastoma 2a (N2a) cells compared to isotonic control cells (310 mOsm) ($n=5$). (B-C) dSTORM imaging showing PLD2 trafficking from ganglioside (GM1) to phosphatidylinositol 4,5-bisphosphate (PIP₂) clusters in response to 70 mOsm stretch in N2a cells (coloring the same as in panel A). Cells were treated, fixed, and labeled with anti-PIP₂ or PLD2 antibody or cholera toxin B (CTxB, GM1). Prior to stretch, PLD2 clustered with GM1 lipids and very little with PIP₂ (grey lines). After stretch PLD2 clustered robustly with PIP₂ but very little with GM1 lipids. Insets show the change in correlation at 5nm, Student's t-test ($n=5-9$), * $p<0.05$, ** $p<0.01$, *** $p<0.001$. (D) Summary figure for the proposed combined model of TREK-1 in a biological membrane by evoked PLD2-depended currents and mechanical activation by direct force from lipid (FFL). TREK-1 in the open conformation (green ions), in complex with PLD2, is shown after stretch (large red arrows) associating (curved black arrow) with PIP₂ clusters (blue shaded bars) in the thin liquid disordered (L_d) region of the membrane (light grey). A known gating helix (grey cylinder) is shown in the up (open channel) position with a PLD2 binding site immediately following the helix (green tube). The opening is a response to three factors that combine to raise the gating helix to the up position. 1) FFL (small red arrows) in TREK-1 favors an open (up) helix conformation. 2) The tip is brought into proximity of PLD2 in the open position and 3) PA (red sphere) is produced and maintains the up positioned by binding to charged residues (blue tube) pulling the helix toward the membrane.

PA regulation of mechanosensitivity thresholds *in vivo*

In order to gain deeper insights into the *in vivo* role of PLD2 in mechanotransduction, we conducted investigations into mechano-thresholds and pain perception in *Drosophila*, employing a PLD-knockout model (Thakur et al., 2016). It is worth noting that, while the specific downstream effectors of PLD in fruit flies remain unidentified, and they possess only distantly related TREK-1 orthologs, (as discussed in a later section), the utilization of fruit flies as a model organism is advantageous because they possess a single PLD gene (*dPLD*).

Our analysis revealed the presence of GM1 domains in dissected fly brains, and we observed that dPLD exhibited a robust response to shear stress of 3 dynes/cm² when cultured in neuronally derived fly cell line, BG2-c2 (see Fig. 6A-B). The activity of PLD increased by nearly fourfold in the fly cell line (Fig. 6A), a result consistent with the response of PLD to mechanical force observed in HEK293T cells (Fig. 3G).

Subsequently, we preceded to examine the *in vivo* role for PA in mechanosensation employing single-animal measurements of arousal threshold (Murphy et al., 2017, 2016) (see Fig. S6A). The arousal assay quantifies the amount of mechanical stimulation required to elicit movement from a resting fly. Over a period of 24 hours, the flies were exposed to a series of incremental vibrational stimuli every 30 minutes. For each series, the threshold of stimulation necessary to induce movement in the fly, as indicated by observable motion, was recorded using automated machine vision tracking. These recorded measurements were then compared to genetically matched control groups.

PLD^{null} flies exhibited a notably lower arousal threshold compared to their control counterparts (Fig. 6C). This lower arousal implies an increased sensitivity to mechanical force. Furthermore, we employed a neuronal-specific driver, Nsyb GAL4, in conjunction with a PLD RNAi line (PLD-KD) to investigate the role of PLD in the central nervous system. The neuronal knockdown of PLD yielded a similar increase in mechanosensitivity, providing clear evidence that this phenotype is specific to neuronal functions (Fig. 6D).

Furthermore, we conducted assessments to determine the role of PLD in fly shock avoidance, serving as a measure of an adverse electrical stimulus. To assess their responses, PLD-KD flies were positioned at the choice-point of a T maze, where they were given the option to select between a chamber inducing a noxious shock or a non-shock control chamber (Drago and Davis, 2016). Flies were subjected to incrementally increasing voltages of electrical shock. Our findings revealed that PLD^{null} flies exhibited heightened sensitivity to electric shock compared to control groups (Fig. 6E-F). In Fig. 6G, we present a potential model illustrating how PA may contribute to mechanosensitivity in a fly.

Discussion

In summary, our findings collectively demonstrate that the spatial distribution of TREK-1 and PLD2 in association with GM1 and PIP₂ lipids, along with the generation of PLD2-derived PA, are essential for eliciting the complete mechanically evoked current from TREK-1 in a biological membrane. Furthermore, the disruption of PLD2 localization within a lipid nanodomain provides a compelling explanation for how the C-terminus confers mechanosensitivity to a channel, particularly when the domain lacks structural integrity.

In contrast, the activation of purified TREK-1 in soy PC does not necessitate the involvement of the C-terminus, as indicated by prior research (Stephen G Brohawn et al., 2014). This observation implies the existence of a PLD2-independent mechanism of action. The precise relative contributions of the two mechanisms in endogenous tissue remain unclear. Notably, in HEK293T

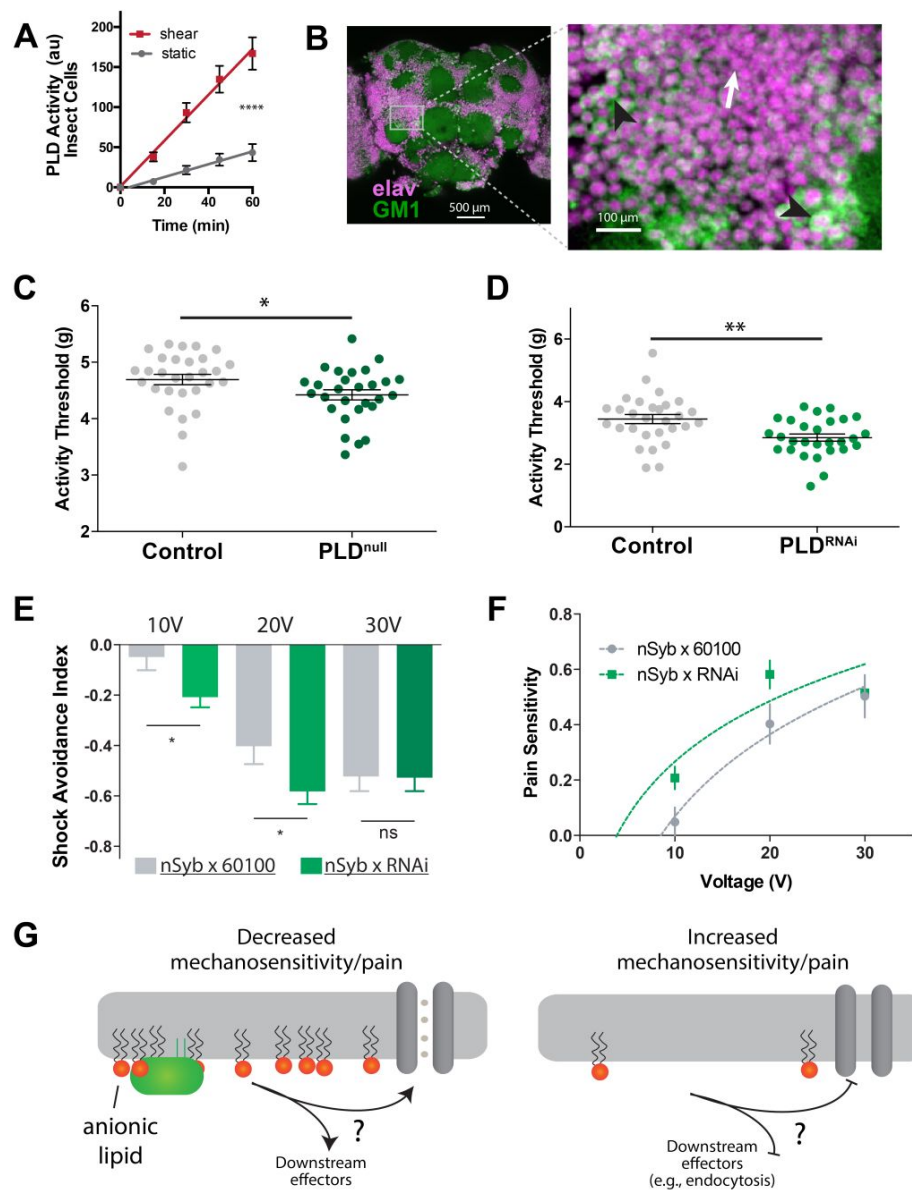


Figure 6.

PLD modulates mechanosensitivity in *Drosophila*.

(A) Shear (3 dynes/cm²) robustly activates PLD2 in a live PLD assay with cultures neuronal insect cells. (B) Cholera toxin B (CTxB) robustly labels GM1 lipids (GM1, green), throughout the brain of *Drosophila*. (left). The zoomed section (right) shows that most of the labeling is found on the membrane. There are notable variations in the amount of CTxB labeling, with some cells expressing GM1 over the entire membrane (black arrows) while others only have labeling in small puncta (white arrow). (C) Measurements of *Drosophila* mechanosensation *in vivo*. Animals with or without the *pld*^{null} gene were stimulated by increasing amounts of mechanical vibration (see [supplemental Fig. S6](#)). Flies lacking PLD2 had a decreased threshold (i.e., more sensitivity to mechanical stimulation) compared to genetically matched controls (w1118) ($p=0.02$, $n=28-29$), consistent with the prediction that PA decreases excitability of nerves. (D) The same result was observed in a *PLD*^{RNAi} line which results in PLD knockdown only in the neurons of *Drosophila* ($p=0.002$, $n=28-29$) Mann-Whitney test. (E-F) Flies were subjected to increasing voltages of electrical shock in a two-choice assay. PLD-KD flies showed an increased sensitivity to shock when compared with wild-type flies. *PLD*^{RNAi} flies had a higher aversion to shock at 10V ($p=0.0213$, $n=21$) and 20V ($p=0.0492$, $n=27-30$), but not at 30V ($p=0.672$, $n=12$). (G) Proposed role of PLD2 in regulating mechanical thresholds. PA is a signaling lipid in the membrane that activates hyperpolarizing channels and transporters. When PA is low the membrane is less polarized, and cells are more sensitive to mechanical activation. The downstream targets are unknown (shown with a '?'). Flies lack a known mechanosensitive TREK-1 homolog.

cells, a current density of 0.05 pA/cm² current density (representing less than 10% of the total stretch current) was found to be PLD2 independent (see **Fig. 1C** [inset](#)). This observation raises the possibility that the localization of TREK-1 to mechanosensitive lipids essential for its activation might be lacking in this context, or alternatively, the propagation of tension within the plasma membrane might differ. Consequently, there is a need for a more comprehensive understanding of TREK-1 spatial patterning within specific cell types, coupled with an understanding of stretch induced tension in the different lipid nanodomains. Furthermore, it is plausible that a portion of the PLD2-dependent stretch current merely augments the direct mechanosensitivity of TREK-1. However, it is important to note that this enhancement mechanism is not obligatory, as the channel is directly activated by PA in purified lipids absent stretch and tension (Cabanos et al., 2017 [inset](#)). This can also explain the discrepancies between liposome-based Mechanosensitivity since crude soy extracts (azolectin) are known to contain anionic lipids, thus may negate the need for a C-terminus and providing the PA needed for full TREK activation.

As mentioned previously, PLD2 is a soluble enzyme that associates with the membrane through palmitate. Given its lack of a transmembrane domain, mechanisms of mechanosensation involving hydrophobic mismatch, tension, midplane bending, and curvature can be largely ruled out, leaving the kinetic mechanism as the most plausible. The palmate moiety establishes van der Waals bonds with the GM1 lipids. The disruption of these bonds is necessary for PLD to disassociate from GM1 lipids, suggesting a mechanism akin to ‘shear thinning’ within GM1 clusters. Shear thinning is a term in rheology that describes the phenomenon wherein viscous mixtures become less viscous in response to shear or stretching forces (Küçükşönmez and Servantie, 2020 [inset](#)). This process is inherently kinetic in nature and operates by mechanically disrupting noncovalent bonds, thereby allowing the molecules to move relative to one another (See **Fig. 3H** [inset](#) and **SSF** [inset](#) for detailed description).

The spatial patterning of TREK-1 with PIP₂ clusters may depend on the channel’s conformational state, as evidenced by the observation that xPLD/TREK-1 combination exhibited reduced associated with PIP₂ after shear compared to the mPLD/TREK-1 combination (**Fig. 2D,F** [inset](#)) despite an increase in the levels of PIP₂ and TREK-1 expression prior to shear (**Fig. S2C-D,G** [inset](#)). This observation aligns with the up down model of the gating helix (Stephen G. Brohawn et al., 2014 [inset](#)). As illustrated in **Fig. 5D** [inset](#), we propose a theoretical mechanism that accounts for the coordinated signaling involving PA and for combined PA signaling and direct force from lipid in a biological membrane. This mechanism suggests that conformational changes conducive to the helix-up (open) position is sustained only when PLD2 is present and capable of producing localized PA. In the absence of PA, PIP₂ forces the channel closed (Cabanos et al., 2017 [inset](#); Chemin et al., 2007b [inset](#)) and the helix in the down position (**Fig. S3E-F** [inset](#)).

In control experiments that estimated protein levels through fluorescent labeling, we observed a significant decrease in TREK-1 protein levels after fixation with mPLD2 or shear for 15 min (**Fig. S2G** [inset](#)). While it is conceivable that this decline in protein levels might have occurred in less than 2 milliseconds, i.e., faster than we are able to observe (see **Fig. S3D** [inset](#)), which could explain the loss of TREK-1 current in **Fig. 1b** [inset](#), independent of local PA gating of TREK-1, endocytosis, the process that removes TREK-1 from the plasma membrane, is not likely to occur in less than 2 ms (Soykan et al., 2017 [inset](#)). Furthermore, the reduction in TREK-1 levels was comparable when wt. PLD2 was employed (**Fig. S2G** [inset](#)), suggesting that alterations in protein levels are unlikely to account for the diminished TREK-1 pressure currents observed with xPLD2 in **Fig. 1B** [inset](#).

It is important to note that PLD2 is recognized for its ability to enhance endocytosis (Du et al., 2004 [inset](#)). Specifically, xPLD2 has been shown to inhibit both agonist induced and constitutive endocytosis of μ -opioid receptor, effectively retaining the receptor on the membrane (Koch et al., 2006 [inset](#)). The phenomenon of membrane retention is not unique to the μ -opioid receptor but extends to many proteins including Rho (Wheeler et al., 2015 [inset](#)), ARF (Rankovic et al., 2009 [inset](#)), and ACE2 (Du et al., 2004 [inset](#)) among others. Consequently, it is not surprising that the overexpression of

xPLD resulted in an increase in TREK-1 surface expression (**Fig. S2C,G**). Conversely, the overexpression of mPLD led to the lowest levels of TREK-1 expression, which aligns with its propensity for promoting endocytosis (**Fig. S2G,H**). Given that mPLD2 yielded the highest mechanically evoked TREK-1 current despite the lowest protein levels, it appears unlikely that protein expression levels are the limiting factor governing mechanically evoked TREK-1 currents in HEK293T cells.

The latency of PLD2 dependent activation is important as it offers insights into the potential physiological processes in which PLD2 may play a role in mechanotransduction. Drawing from estimates involving diffusion from GM1 to PIP₂, we estimated a latency of ~650 μ s (Petersen et al., 2016). To establish an upper temporal limit for PLD2 to become activated and generate PA in proximity to TREK-1, we leveraged the rate of increase in TREK-1 activity observed with WT PLD2. We observed initial TREK-1 currents almost immediately and these currents became notably significant within 2.1 ms at a pressure of 60 mmHg (**Fig. S3D**). This timeframe is highly likely to be faster than the margin of error associated with our instrument setup. While we did not perform a precise calibration of our setup's error, conservative estimates based on manufacturer specifications suggest an upper limit of approximately 10 milliseconds.

The requirement for both PLD2 activity and the C-terminus to activate TREK-1 by pressure provides further validation for the previously proposed conclusion that TREK-1 is gated by a local high concentration of PA (Cabanos et al., 2017; Comoglio et al., 2014). In theory, PLD2 activity could elevate global PA levels to a point where TREK-1 becomes activated without the need for spatial patterning through protein-protein interactions. However, our experiments with truncated TREK-1 and 60 mmHg pressure in HEK293T cells did not support this hypothesis (**Fig. 1C**).

Our findings regarding the mechanism we identified remained consistent across cell types (HEK, N2a, C2C12, mouse neurons) despite their significant differences in cellular characteristics. We were able to induce protein-lipid domain associations in all these cell types through the manipulation of cholesterol levels. This intriguing observation suggests a potential avenue of research to investigate whether cholesterol or other lipids undergo significant changes at sites where mechanosensation is known to occur, such as at the termini of low- or high-threshold mechanoreceptors (Handler and Ginty, 2021). It's conceivable that various neuronal types and the channels localized within them adapt to different types of mechanosensation by modulating the lipid environment where sensory perception is most critical. For instance, inflammation-induced cholesterol uptake (Hansen and Wang, 2023) might sensitize channels to pain perception (Petersen et al., 2020).

Additionally, our experiments with flies highlight the importance of the upstream effectors in these systems. While the specific binding partners for PLD2 and its potential downstream effects in flies remain unknown, the observed phenotypes strongly suggest that PLD plays a crucial role in normal sensory perception which appears to be more evolutionarily conserved than TREK-1. It is possible that a single lipid or a mechanism regulating multiple channels simultaneously may have a more significant impact compared to the modulation of a single channel.

The regulation of mechanosensation and electric shock responses in *Drosophila* by PA (**Fig. 6**) provides *in vivo* support for the growing understanding of the role of anionic lipids in setting force-sensing thresholds. For example, PIP₂ sets the threshold for mechanical B-cell activation (Wan et al., 2018), while sphingosine-1-phosphate (S1P), a lipid similar to PA, regulates pain in mice (Hill et al., 2018). The activation of PLD2 by mechanical force and substrate presentation helps elucidate how an enzyme could directly set pain thresholds and mechanosensitivity via canonical mechanosensitive ion channels and endocytosis.

Palmitoylation, a modification found in many important signaling molecules including tyrosine kinases, GTPases, CD4/8, and almost all G-protein alpha subunits (Aicart-Ramos et al., 2011 [DOI](#)), can target proteins to GM1 domains (Levental et al., 2010 [DOI](#)). The spatial patterning of these palmitoylated proteins or their binding partner, by mechanical force may alter the availability of substrates and affect downstream signaling. Spatial patterning along with its effect on palmitoylated G-proteins likely contributes to the mechanosensitivity observed in many GPCRs (Gudi et al., 1998 [DOI](#); Storch et al., 2012 [DOI](#); Wei et al., 2018 [DOI](#); Xu et al., 2018 [DOI](#)). Many ion channels are also palmitoylated (Shipston, 2011 [DOI](#)). For example, the voltage-gated sodium channel (Na_v)1.9 clusters with GM1 lipids, and disrupting this clustering (e.g., by chemically removing cholesterol) induces a corresponding pain response (Amsalem et al., 2018 [DOI](#); Wan et al., 2018 [DOI](#)). Similarly, sphingosine-1-phosphate (S1P), an anionic lipid similar to PA, regulates pain in mice (Hill et al., 2018 [DOI](#)).

Lastly, it's worth noting that PIP2 gates many ion channels (Cabanos et al., 2017 [DOI](#); Chung et al., 2019 [DOI](#); Hansen, 2015 [DOI](#); Hansen et al., 2011 [DOI](#); Robinson et al., 2019 [DOI](#)), and the spatial patterning induced by PIP2 should be considered in combination with the direct effect of the lipid on the channel. Cholesterol, known to bind directly to and modulate channels (Levitan et al., 2014 [DOI](#)), should also be considered separately from its influence on spatial patterning through GM1 lipids. These direct lipid effects are likely to interact with spatial patterning to collectively regulate the biochemical processes within a cell.

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Author Contributions

Conceptualization, E.N.P. and S.B.H.; Methodology, E.N.P., M.G., K.R.M., and S.B.H.; Investigation, E.N.P., M.G., and K.R.M.; Resources, W.W.J., E.M.J., and S.B.H.; Writing – Original Draft, E.N.P. and S.B.H.; Writing – Review and Editing, E.N.P., M.G., M.A.P., E.M.J., and S.B.H.; Supervision, W.W.J. and S.B.H.; Funding Acquisition, W.W.J. and S.B.H.

Materials and Methods

Cell Culture and Gene Expression

HEK293T cells (ATCC Cat# CRL-3216, RRID:CVCL_0063) were cultured in DMEM (Corning cellgro) with 10% FBS, 100 units/mL penicillin, and 100 µg/mL streptomycin. Cells were plated on poly-D-lysine-coated 12 mm microscope cover glass at approximately 12h, 36h, or 60 hours before transfection with genes for target proteins. Transfections were performed using X-tremeGENE 9 DNA transfection agent (Roche Diagnostics). Full-length human TREK-1 (TREK-1 FL) with C-terminus GFP tag in pCEH vector was obtained from Dr. Stephen Long. Mouse PLD2 constructs (mPLD2) and inactive mutant (K758R single mutation, xPLD2) without GFP tag in pCGN vector

were provided by Dr. Michael Frohman. TREK-1 constructs were co-transfected with mPLD2 or xPLD2 at a 1:4 ratio (0.5g of TREK-1 and 2g of PLD DNA)(Comoglio et al., 2014 [Comoglio et al., 2014](#)). All the salts for internal/external solutions were purchased from either Sigma or Fisher Scientific.

Electrophysiology

The transfected HEK293T cells were used in 24~36 hours after transfection for standard excised inside-out patch-clamp recordings of TREK-1(Stephen G Brohawn et al., 2014 [Brohawn et al., 2014](#); Comoglio et al., 2014 [Comoglio et al., 2014](#)). Currents were recorded at room temperature using an Axopatch 200B amplifier and Digidata 1440A (Molecular Devices). Borosilicate glass electrode pipettes (B150-86-10, Sutter Instrument) were pulled with the Flaming/Brown micropipette puller (Model P-1000, Sutter instrument) resulting in 3~6 MΩ resistances with the pipette solution (in mM): 140 NaCl, 5 KCl, 1 CaCl₂, 3 MgCl₂, 10 TEA-Cl, 10 HEPES, pH 7.4 (adjusted with NaOH). Bath solution consists of (in mM): 140 KCl, 3 MgCl₂, 5 EGTA, 10 TEA-Cl, 10 HEPES, pH 7.2 (adjusted with KOH). Low concentration of TEA (10 mM), was added into both pipette/bath solutions to block the endogenous potassium channels in HEK293T cells. Patch electrodes were wrapped with parafilm to reduce capacitance. Currents measured using Clampex 10.3(Molecular Devices) were filtered at 1 kHz, sampled at 20 kHz, and stored on a hard disk for later analysis.

Pressure clamping on the patch was performed using high speed pressure clamping system (ALA Scientific) through the Clampex control. Data was analyzed off-line by a homemade procedure using IgorPro 6.34A (WaveMetrics).

TREK-1 current, either co-expressed with mPLD2 or xPLD2 was activated by negative pressure steps from 60 to 0 mmHg in 10 mmHg decrements at +30 mV membrane potential, and 5 traces for each case were recorded and averaged for the analysis. Patch size was estimated using the current density (I_{density} ; pA/ μm^2). Then, a Boltzman equation, $I_{\text{density}} = \text{base} + \frac{\text{max} - \text{base}}{1 + \exp((P50 - P)/\text{slope})}$ was used to fit the data with a constraint of base=1 due to poor saturation of the current at high pressure. P is the applied pressure, P50 is the pressure that activates 50% of maximum current measured (non-saturating), and slope shows the sensitivity of current activation by pressure. In some experiments with hTREK-1+xPLD2 co-expression where the activated currents were too small to fit to the Boltzman equation, the current amplitude at P=-30 mmHg (I_{m30}) was compared with its 5x standard deviation(I_{5xSD}). If $I_{\text{m30}} < I_{\text{5xSD}}$, the experiment was excluded from the Boltzman equation fitting and corresponding P50-slope analysis. This empirical rule (we call it 5xSD rule) can discern 4 out of 5 wild type (TREK-1 FL) cell-attached recording cases as null experiments suggesting that it could be a usable/useful empirical criterion for our experiment. Then, the current density at -60 mmHg and P50-slope data were used for statistical analysis. Mann Whitney test was done to assess statistical significance using Prism6 (GraphPad software), and outliers were eliminated using a built-in function in Prism with Q = 1 %. The values represented are Mean +/- SEM.

Fixed cell preparation

Cells (C2C12, HEK293T and N2a) were grown to 80% confluence (C2C12 were allowed to differentiate overnight in serum free media). Cells were rinsed, treated as needed, and then fixed with 3% paraformaldehyde and 0.1% glutaraldehyde for 10 min to fix both protein and lipids. Glutaraldehyde was reduced with 0.1% NaBH₄ for 7 min followed by three-10 min washes with PBS. Cells were permeabilized for 15 min with 0.2% Triton X-100, blocked with 10% BSA/0.05% Triton/PBS at rt. for 90 min. Primary antibody (PLD2, Cell Signaling #13891; TREK-1, Santa Cruz #sc-50412; PIP₂, Echelon Biosciences #z-P045) was added to a solution of 5% BSA/0.05% Triton/PBS for 60 min at rt at a concentration of 1:100 followed by 5 washes with 1%BSA/0.05% Triton/PBS for 15 min each. Secondary antibody (Life Technologies #A21244 and A21235; cy3B antibodies were produced as described previously(Petersen et al., 2016 [Petersen et al., 2016](#))) was added in the same buffer as primary for 30 min at rt followed by 5 washes as above. A single 5 min wash with PBS was

followed by a post-fix with fixing mixture, as above, for 10 min w/o shaking. This was followed by three-5 min. washes with PBS and two-3 min. washes with dH₂O. Cells only receiving CTxB treatment were not permeabilized.

The dual-fixation protocol is used to minimize any effects from post-fixation aberrations. While always good practice for super-resolution in general, this dual fixation also ensures that the movement of any molecule of interest which may not have been immobilized by the initial fixation can be fully immobilized after labeling since the antibodies or toxins used for labeling will be efficiently cross-linked during this post-labeling fixation step. While some have proposed that this problem should be solved by adding the label before the initial fixation (Tanaka et al., 2010) we believe that in the absence of easily-attainable monomeric labeling molecules it would have likely led to clustering artifacts due to the (often) multimeric nature of the labeling proteins.

For cells loaded with cholesterol, 4 ug/mL apolipoprotein E3 (BioLegend, USA) was mixed with fresh 10% FBS and applied to the cells for 1hr prior to shear and or fixing.

Shear force was applied to cells in ibidi μ -Slide I^{0.4} Luer chambers with a flow rate calibrated to apply 3.0 dynes/cm². Fixation media (see above) was applied to cells using a syringe pump (Harvard Apparatus PHD ULTRA) and kept at 37°C using an in-line heater (Warner SH-27B).

TREK-1 in HEK293T cells was labeled with a EGFP concatenated to the C-Terminus of full-length human TREK-1 (Fig. S3b), or by applying anti-TREK-1 antibody (sc-398449, Santa Cruz) conjugated to cy3b (Petersen et al., 2016). Anti-PIP₂ antibody was directly conjugate alexa 647 using the same protocol.

Super-resolution dSTORM

Images were recorded with a Vutara 352 and VXL super-resolution microscopes (Bruker Nano Surfaces, Salt Lake City, UT) which is based on the 3D Biplane approach. Super-resolution images were captured using a Hamamatsu ORCA Flash4.0 sCMOS camera and a 60x water objective with numerical aperture 1.2. Data were analyzed by the Vutara SRX software (version 5.21.13 for the data in Figures 3–5 and version 7.0.07 for the data in Figure 2). Single molecules were identified by their brightness frame by frame after removing the background. Identified particles were then localized in three dimensions by fitting the raw data in a customizable region of interest (typically 16 X16 pixels) centered on each particle in each plane with a 3D model function that was obtained from recorded bead data sets. Fit results were stored as data lists for further analysis.

Fixed samples were imaged using a 647 nm and 561 nm excitation lasers, respectively, and 405 nm activation laser in photo switching buffer comprising of 20 mM cysteamine (Sigma, #30070), 1% betamercaptoethanol (BME) (Sigma, #63689) and oxygen scavengers (glucose oxidase, GLOX) (Sigma #G2133) and catalase (Sigma #C40)) in 50mM Tris (Affymetrix, #22638100)+10 mM NaCl (Sigma, #S7653) +10% glucose (Sigma, #G8270) at pH 8.0 at 50 Hz and maximal powers of 647 nm, 561 nm and 405 lasers set to 8, 10, and 0.05 kW cm⁻² respectively.

Three-color EGFP-STORM. The dSTORM with EGFP (EGFP-STORM) was performed identical to the two-color dSTORM described above except that a 488 laser was also used to excite EGFP and PIP₂ and CTxB were directly conjugated with fluorescent dyes (Atto 647 and cy3b 555 respectively) obviating the need for fluorescent secondary antibodies. TREK-1 and PIP₂ antibodies were conjugated with NHS esters of either Cy3B or Atto 647. 1.5 mg of antibody was conjugated to 3 ng of dye in 1M NaHCO₃ buffer pH 8 for 1 hour at room temperature and separate on a NAP-5 desalting column. The acquisition was performed with no 405-activation laser in GLOX/BME buffer. The GLOX/BME buffer was not required for EGFP blinking, but it did improve the fluorescence and the number of localization particles determined. The resolution TREK-1 localizations determined by EGFP_TREK-1 and cy3b labeled anti TREK-1 antibody were comparable (45-50 nm resolution).

The pair correlation function $g(r)$ and cluster analysis were performed using the Statistical Analysis package in the Vutara SRX software. Pair correlation analysis is a statistical method used to determine the strength of correlation between two objects by counting the number of points of probe 2 within a certain donut-radius of each point of probe 1. This allows for localization to be determined without overlapping pixels as done in traditional diffraction-limited microscopy. For three-color EGFP-STORM probes 1 and 3 and 2 and 3 were also compared using the pair correlation function. Localization at super resolution is beyond techniques appropriate for diffraction-limited microscopy such as Pearson's correlation coefficient. Lipid cluster size was determined using the DBSCAN clustering algorithm also included as part of the Vutara SRX software.

Fluorescence Recovery After Photobleaching (FRAP)

C2C12 cells were grown in DMEM with 10% FBS until 16 hours before use in which they were switched into serum free DMEM. On the day of the experiment, DMEM in live cells was replaced with DMEM w/o phenol red. Fixed cells were rinsed once with PBS and then put into a mixture of PBS with 3% PFA and 0.1% glutaraldehyde for 20 min at 37°C. Fixed cells were then rinsed with PBS 5 x 5 min and then placed back into phenol-free DMEM. CTxB (ThermoFisher C34778, 100 µg/ml) was then applied 1:200 into each plate and allowed to incubate for >30 min before imaging. Imaging and data collection was performed on a Leica SP8 confocal microscope with the Application Suite X v.1.1.0.12420. 5 images were taken as baseline after which a selection of 1 or more ROI was bleached at 100% laser power for 6-8 frames. Recovery was measured out to 5 min and fluorescence of the ROI(s) were quantified. The fluorescence before bleaching was normalized to 1 and after the bleaching step was normalized to 0.

Cholesterol Assay

N2a cells were cultured in 48 well plates with 200µL media in each well and then changed to 200µL PBS for the shear treatment. The shear plate was incubated with PBS on an orbital rotator at 3dyn/cm² for 10 min in a 37°C incubator. The control plate was incubated with PBS for 10 min in the same incubator with no shear. Then the shear plate was incubated with 200µL 4%PFA+0.1% glutaraldehyde in PBS for 10 min with 3dyn/cm² shear and 10 min without shear. The control plate was fixed for 20 min with no shear. Fixed cells were lysed in RIPA buffer (Thermo) and assay in 100 µL PBS with, 4 U/mL cholesterol oxidase, 100 µM amplex red, and 2 U/ml horseradish peroxidase (HRP).

In vitro cellular PLD Assay

In vitro cellular PLD2 activity was measured in cultured HEK293T cells by an enzyme-coupled product release assay(Petersen et al., 2016 [DOI](#)) using amplex red reagent. Cells were seeded into 96-well plates (~5×10⁴ cells per well) and incubated at 37 °C overnight to reach confluency. The cells were starved with serum-free DMEM for a day and washed once with PBS (phosphate-buffered saline). The PLD reaction was initiated by adding 100 µL of reaction buffer (100 µM amplex red, 2 U/ml horseradish peroxidase (HRP), 0.2 U/ml choline oxidase, and 60 µM C8-PC in PBS). The assay reaction was performed for 2-4 hour at 37 °C and the activity was kinetically measured with a fluorescence microplate reader (Tecan Infinite 200 Pro) at excitation and emission wavelengths of 530 nm and 585 nm, respectively. The PLD2 activity was calculated by subtracting the background activity (reaction buffer, but no cells). For the bar graphs, samples were normalized to the control activity at the 120 min time point.

Drosophila Assays

For behavior experiments, 1 to 5-day old flies were collected in vials containing ~50 flies at least 12 hours before the experiment. Flies were allowed to acclimate to behavior room conditions for >30 min (dim red light, ~75% humidity) before each assay. Shock avoidance was tested by placing flies in a T-maze where they could choose between an arm shocking at the indicated voltage every

2 seconds and an arm without shock. Flies were given 2 min to choose which arm, after which flies were collected and counted to determine the shock avoidance index for each voltage and genotype. Control and knockout flies were alternated to avoid any preference and the arm used for shock was also alternated to control for any non-shock preference in the T-maze itself. Shock avoidance index (AI) was calculated as $AI = (\# \text{ flies shock arm} - \# \text{ flies control arm}) / \# \text{ flies total}$. Plotting the inverse of this metric we obtain a pain sensitivity curve in which we observe a right-shift when the *pld* gene was knocked down (**Fig. 3f**).

Arousal threshold protocol has been described in detail previously (Murphy et al., 2016). Briefly, animals were exposed hourly to a series of vibrations of increasing intensity ranging from 0.8 to 3.2 g, in steps of 0.6 g. Stimuli trains were composed of 200 ms vibration with 800 ms inter-vibration interval and 15 s inter-stimuli train interval. Stimulation intensity and timing were controlled using pulse-width modulation via an Arduino UNO and shaftless vibrating motors (Precision Microdrives, model 312–110). Arousal to a given stimulus was assigned when an animal (1) was inactive at the time of the stimulus, (2) satisfied a given inactivity criteria at the time of the stimulus, and (3) moved within the inter-stimuli train period (15 s) of that stimulus.

Statistics

All statistical calculations were performed using a student's t-test or Mann-Whitney test in Prism software (v9) unless otherwise noted. For statistics of more than one point along a pair correlation curve a nested Students t-test was used. Significance is noted as follows: ns, $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$.

Supplemental Discussion

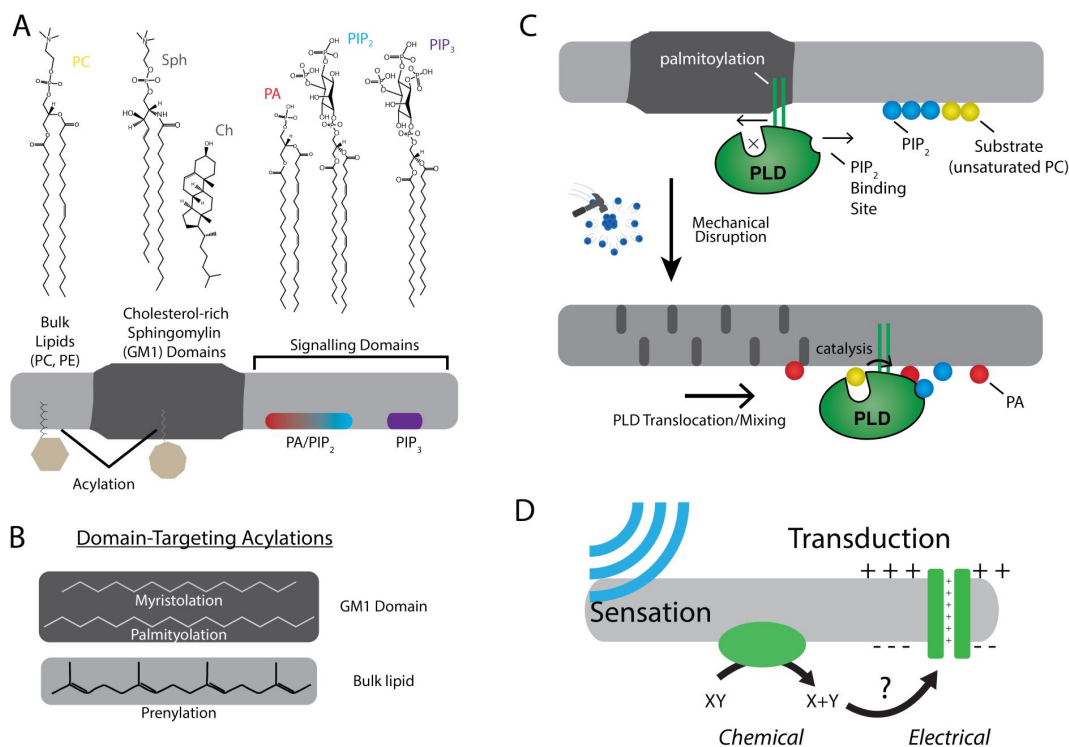
In our study, we utilized cholera toxin B (CTxB) and antibody probes for lipid labeling, recognizing that these lipid probes have previously been shown to induce artificial clustering in unfixed cells a phenomenon referred to as “antibody patching” (Moon et al., 2017; Petersen et al., 2020; Veatch et al., 2012). Antibody patching has historically served as a useful tool to segregate proteins within biological membrane, facilitating the determination of their precise localization (Lang et al., 2001; Yuan and Hansen, 2023). In the context of our research, we anticipate that antibody patching might similarly affect lipid distribution, offering a valuable advantage in clearly elucidating the co-localization of proteins with lipids in the membrane.

To address concerns related to potential artifacts stemming from clustering, we adopted a two-step fixation procedure. Initially, cells were fixed during treatment to minimize lipid mobility, followed by a second fixation step after labeling, ensuring fixed antibodies and high localization precision during dSTORM. Additionally, we assessed lipid diffusion after fixing, using our typical experimental protocols. In our system, combined paraformaldehyde and glutaraldehyde effectively inhibited GM1 lipid diffusion, as confirmed by fluorescence recovery after photobleaching (FRAP) experiments conducted in live cells (**Fig. S5g-h**).

It is also worth considering potential artifacts arising from variations in labeling density and overcounting, which could impact the size of the lipid clusters analyzed in **Figure 3**. We quantified the amount of fluorescent labeling (**Fig. S5h**) and observed a slight reduction in GM1 and PIP₂ labeling after shear, although this change did not reach statistical significance.

The mechanical disruption of PLD2 with GM1 lipids and its association with PIP₂ was determined by the pair correlation function, a method that is robust against artifacts associated with changes in labeling density (Coltharp et al., 2014). Consequently, our data indicating the movement of proteins between domains remains largely independent of potential artifacts stemming from

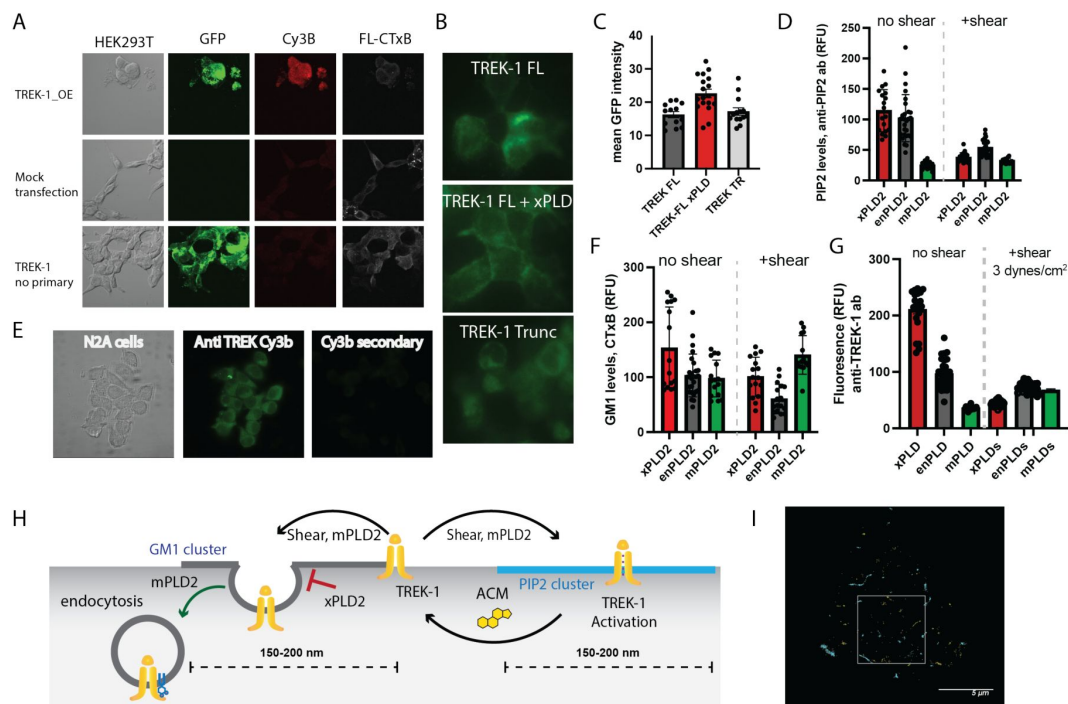
artificial clustering(Moon et al., 2017 [↗](#); Petersen et al., 2020 [↗](#); Robinson et al., 2019 [↗](#)). This conclusion is further supported by the consistency of dSTORM results obtained using both EGFP and cy3b-anti-TREK-1 antibody (**Fig. 2** [↗](#), **S4** [↗](#)).



Supplemental Figure 1.

The role of lipids and lipid order in mechano-transduction.

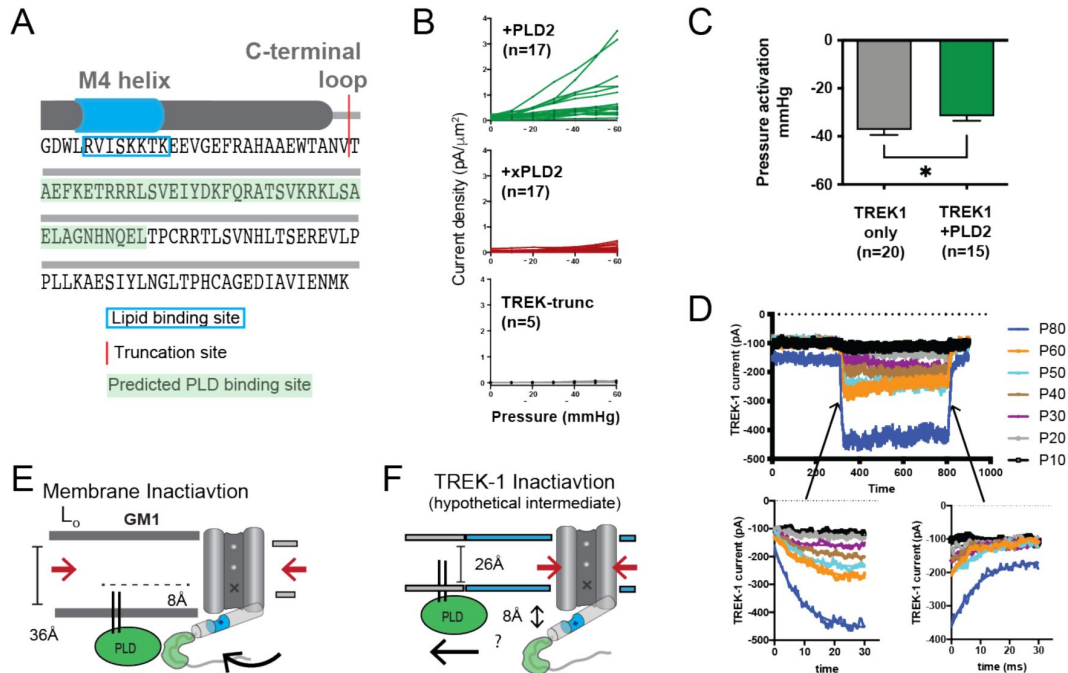
(A) The plasma membrane is composed of lipids which can cluster into separate and distinct domains with unique properties such as thickness and charge. Domains for saturated gangliosides (GM1) are shown separate from phosphatidylinositol 4,5-bisphosphate (PIP₂) and phosphatidylinositol 3,4,5 triphosphate (PIP₃). These domains contain proteins which are targeted to the domain through post-translational acylation. (B) The types of acylation are shown along with their targeting location. (C, top) The enzyme phospholipase D2 (PLD2, green) is shown with its acylation that binds to the saturated ordered site in GM1. The site discriminates palmitoylated proteins from prenylated proteins. This is called the anesthetic/palmitate (AP) site because anesthetics also compete for this site (not shown) (Pavel et al., 2020; Petersen et al., 2020). (C, bottom) Upon chemical or mechanical disruption of the domain, the binding site is disrupted releasing PLD2 from the AP site. PLD2 is then free to bind PIP₂ where it has access to its substrate phosphatidylcholine (PC). (D) Cartoon depicting a mechanically evoked current through a chemical intermediate.



Supplemental Figure 2.

Expression and staining of TREK-1 in cell culture.

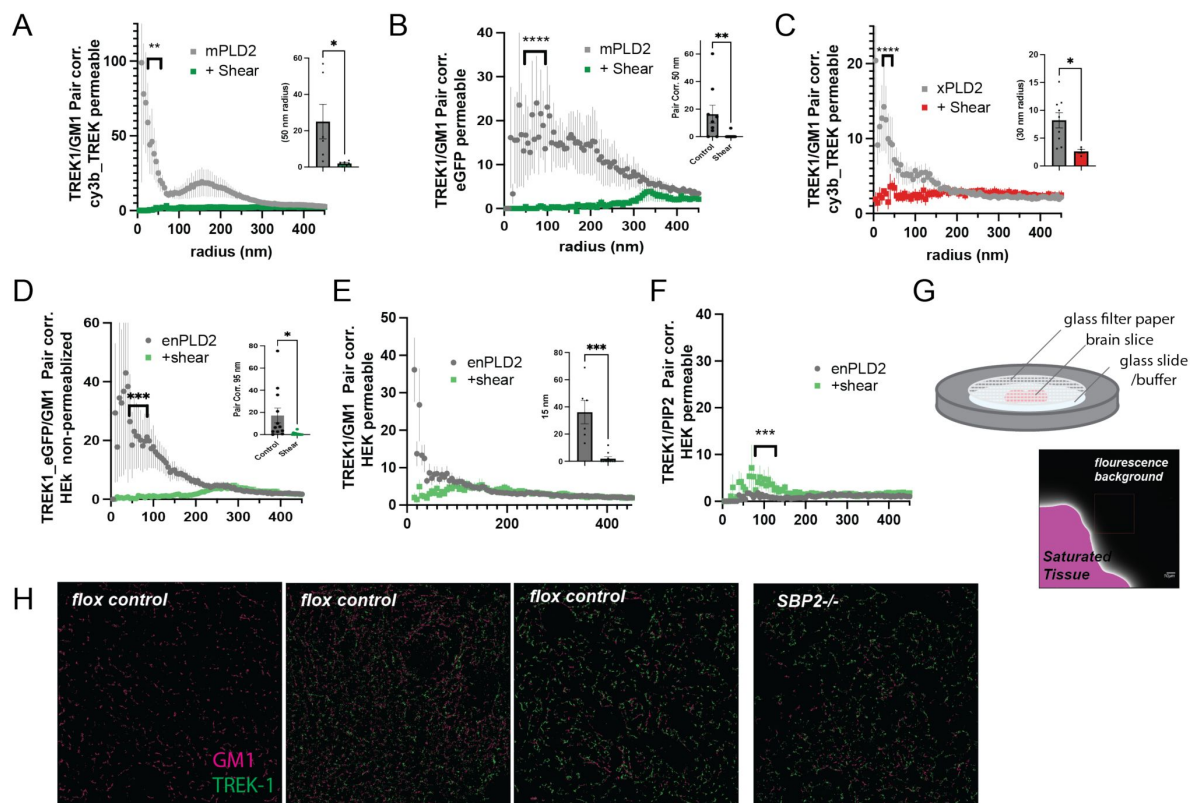
(A) Confocal images of HEK293T cells overexpressing human TREK-1 tagged with EGFP. The cells were fixed and stained with anti-TREK antibody, a secondary cy3b antibody (red) and A647 conjugated cholesterol toxin B (CTxB, grey) to label lipids. In the absence of transfection, no GFP signal is observed. A negative control, lacking the primary antibody, exhibits no fluorescence. (B) Visualization of TREK-1 overexpression in HEK293T cells prepared identical to the cells used for ecophysiology experiments in Fig. (C) Overexpression of catalytically inactive PLD2 (xPLD), appears to enhance TREK-1 expression. Truncating the c-terminus (TREK trunc) has no discernible effect on expression TREK-1 at the plasma membrane. (D) Shear and overexpression of mPLD2 reduce PIP2 levels on the plasma membrane. (E) Immunostaining of endogenous TREK-1 in neuroblastoma 2a (N2a) cells. (F) Both mPLD2 overexpression and shear dramatically reduced the amount of GM1 observed by fluorescent staining with cholera toxin B (CTxB). (G) Assessment of the impact of shear on TREK-1 levels in HEK293T cells. These cells were transfected with full-length TREK-1 (TREK FL) and either xPLD2 or mPLD2. After 24 hours, the cells were subjected to 3 dynes/cm² rotary shear and fixed. Both PLD2 overexpression and shear resulted in a reduction of TREK-1 expression, likely due to endocytosis processes. (H) Cartoon illustrating the role of shear, PLD2, xPLD, in endocytosis. Shear and mPLD2 both activate TREK-1 and decrease TREK-1 surface levels. Presumably the decrease is due to endocytosis since mPLD2 and shear are known to increase endocytosis and xPLD2 is known to block endocytosis. (I) dSTORM images showcasing sheared HEK293T cells expressing TREK-1. The image corresponds to the cell depicted in the main Figure 2b. The highlighted box represents an expanded region of this image.



Supplemental Figure 3.

Electrophysiology details and methods.

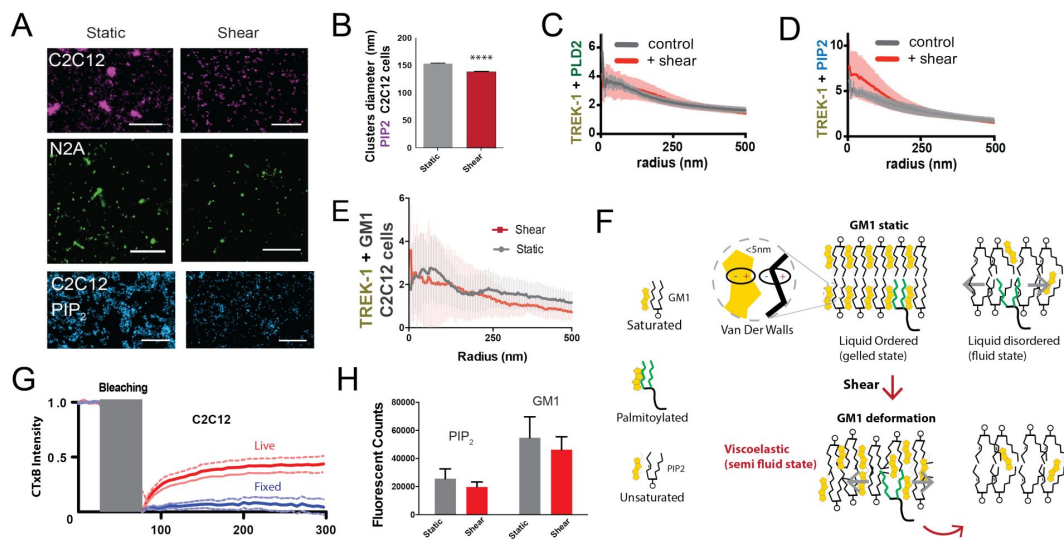
(A) Illustration depicting the C-terminal end of TREK-1. The red region indicates the truncation site used, and the predicted PLD binding site for PLD2 is highlighted. The last transmembrane helix (M4) is depicted as a grey cylinder, and the anionic lipid binding site is highlighted in blue. (B) Individual cell traces showing current densities ($\text{pA}/\mu\text{m}^2$) for TREK-1 co-expressed with PLD2 (green), TREK-1 coexpressed with xPLD2 (red), and TREK-1 with a C-terminal truncation (TREK trunc, grey). (C) Half maximal TREK-1 pressure current within a non-saturating pressure range of 0-60 mmHg. Overexpression of PLD2 significantly reduces the apparent pressure required to activate TREK-1 ($p < 0.05$, $n = 15-20$). (D) Representative cell recording displaying TREK-1 pressure currents in response to pressure steps from 0 to 60 mmHg, taken in 10 mmHg increments. The bottom left panel illustrates the activation step and the bottom right panel shows the deactivation. Both activation and deactivation processes appear to occur within sub-5 ms time frames, near the limit of detection for the experimental setup. (E) Membrane inactivation process. After mechanical stretching, the membrane relaxes, allowing the palmitates from PLD2, to re-associate with the GM1 lipids. Consequently, TREK is drawn into GM1 clusters through its interaction with PLD2. In the absence of phosphatidic acid (PA) and due to an increased hydrophobic thickness of the membrane, the channel's gate assumes the down (closed) position, marked with an "X". (F) Direct inactivation of TREK-1 through an intermediate. Upon reversal of mechanical stretch (relaxation of the membrane), the channel may transition into a closed conformation due to direct pressure exerted on the channel (indicated by the large red arrows). In a thin membrane this action could displace the gating helix up to 8 Å away from the membrane, disrupting the PLD2/TREK-1 interaction. This putative intermediate state is expected to be transient as TREK-1 would likely re-associate in thicker lipid regions.



Supplemental Figure 4.

Comparisons of labeling type and permeabilization.

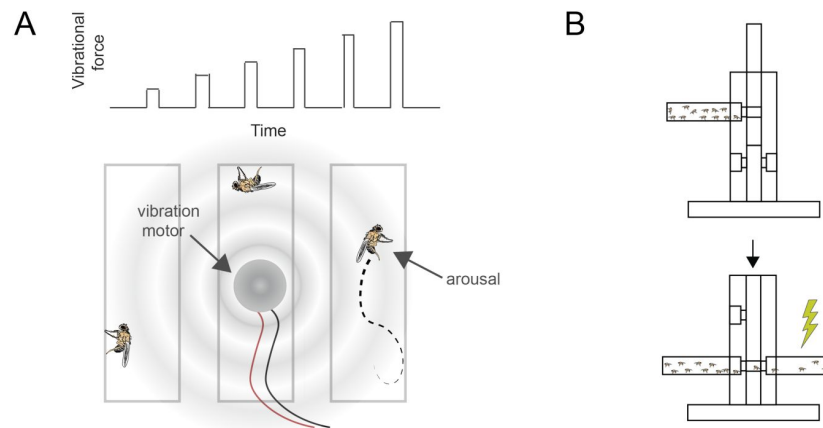
(A) Two-color dSTORM using a Cy3b-labeled TREK-1 antibody after mechanical shear in the presence of over expressed mouse PLD2 (mPLD2) and full-length human TREK-1. HEK293T cells were fixed, permeabilized, and stained with a Cy3b-conjugated anti-TREK-1 antibody. Pair correlation (Pair corr.) analysis of TREK-1 with alexa-647-conjugated cholera toxin B (CTxB) was conducted via dSTORM, revealing a decrease in Pair corr. with 3 dynes/cm² orbital fluid shear. (B) Similar experiments as in panel A, but with Pair corr. analysis from EGFP, which was C-terminally expressed with TREK-1 instead of using a Cy3b-labeled anti-TREK-1 antibody. The use of EGFP in dSTORM buffers resulted in a robust dSTORM signal. This dataset also offers a direct comparison between permeabilized and non-permeabilized cells in [Fig. 2B](#). Pair correlation of TREK-1 with GM1 showed dramatic decreased under three different conditions. (C) A comparison of Cy3b-STORM with EGFP-STORM as observed in [Figure 2E](#) in the presence of xPLD2. (D-E) A comparison between non-permeabilized (a) and permeabilized (b) HEK293T cells over expressing full-length human TREK-1 (TREK FL) and endogenous PLD2 (enPLD2), i.e., without PLD2 overexpression. Taken from ([Call et al., 2023](#)). (F) Pair correlation between TREK-1 and PIP₂ increased slightly after mechanical shear. Insets in (a-d) illustrate the variability at a single radius. Statistical analysis for single points was conducted using a Student's t-test, while statistical comparisons at multiple radii were performed with a nested Student's t-test. (G) Schematic representation of brain slices prepared for dSTORM. Mouse brains, fixed through whole-body perfusion, were sliced and labeled with Cy3b-anti-TREK-1 antibody and A647 CTxB. These slices were mounted on a cover slip with fiberglass filter paper on top to secure the tissue during imaging with dSTORM buffer added to the filter paper. The fluorescent background (640 nm) was undetectable even at saturating light intensities. (H) Example images of brain slices from control and astrocyte specific SREBP2 null mice. TREK-1 was expressed in most brain regions but not uniformly (left panel). Pair correlation analysis was performed on regions with both TREK-1 and CTxB labeling. The displayed slice is a coronal section near the hippocampus, although the precise region of interest is unspecified. Significance was determined by a Student's t-test for a single point. For multiple points a nested student's t test was used (*p<0.05, ** p<0.01, *** p<0.001, ****p<0.0001).



Supplemental Figure 5.

Observing cellular changes in response to mechanical stimulation.

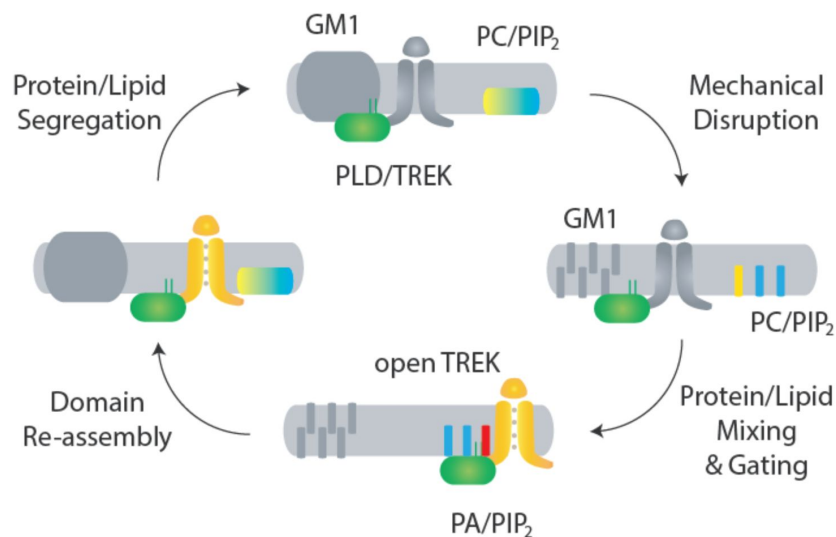
(A) Representative images illustrating the impact of shear on the apparent size of GM1 (maroon and green) in muscle C2C12 and neuronal N2a cells respectively. PIP₂ clusters in C2C12 cells are shown in blue. When shear force is applied the apparent size decreases. Scale bars = 1 μ m. (B) Cluster analysis of PIP₂ demonstrates a small but statistically significant decrease in size. (C) Complex formation between TREK-1 and PLD2 before and after shear in C2C12 cells. Pair correlations analysis of TREK-1 and PLD2 before (grey line) and after (red line) 3 dynes/cm² shear. Their association remains almost identical in both states. (D) Staining of TREK-1 with phosphatidylinositol 4,5 bisphosphate (PIP₂) antibody in C2C12 cells. Before shear (grey line) TREK-1 exhibits significant association with PIP₂. After shear, this association further enhanced, suggesting some TREK-1 complexes move to PIP₂ clusters in C2C12 cells. (E) Pair corr. of TREK-1 with GM1 clusters (CTxB) measured by dSTORM in C2C12 cells prior to cholesterol loading, as shown in Fig. 4b. In the low cholesterol state, endogenous TREK-1 shows minimal association with GM1 lipids. (F) Shear thinning model for PLD2. Cholesterol is depicted as packing with saturated lipids and saturated palmitate via Van der Waals interactions (within 5 nm). Unsaturated lipids contain a double bond that alters the packing surface of a lipid. In a perfectly ordered state, palmitoylated proteins are ordered with the GM1 lipids and cholesterol. In the disordered region palmitates move fluidly within the membrane. After shear the GM1 lipids remain ordered but deformed. The palmitates no longer efficiently pack with the GM1 lipids, reducing their affinity for the ordered domain and allowing the palmitates to move in the membrane. (G) Fluorescence recovery after photobleaching (FRAP) imaging to examine potential labeling artifacts (Moon et al., 2017; Raghunathan and Kenworthy, 2018; Wang et al., 2018) of pentavalent CTxB that might persist after fixation. The duration of photo bleaching is indicated in grey. The blue line shows that fixation with paraformaldehyde (PFA) and glutaraldehyde effectively restricts the large-scale movement of lipids in fixed cells compared to live cells (red). (H) For both PIP₂ and GM1 labeling, shear did not significantly decrease the overall counts measured with dSTORM in C2C12 cells ($p > 0.05$, Student's t-test).



Supplemental Figure 6.

Pan-neuronal knockdown of *pld* in *Drosophila* alters sensitivity to electrical shock.

(A) Demonstrates that PLD activity in neuroblastoma 2a (N2a) cells increases in response to mechanical force. (B) Illustration of the experimental setup for applying mechanical stimulation to *Drosophila* flies. A series of six increasing vibrations (top) used to stimulate the flies. A vibration motor was attached to the back of the chamber containing the flies, and their responses were monitored using a web camera to assess stimulation-induced arousal (bottom).



Supplemental Figure 7.

Proposed model for domain-mediated mechanosensation of TREK-1 by spatial patterning.

(top) In domain-mediated mechanosensation, in an unstimulated state, the palmitate site within GM1 domains is intact and binds tightly to palmitates. This tight affinity sequesters the PLD2/TREK-1 complex away from its activating lipids phosphatidylcholine (PC, yellow) and phosphatidylinositol 4,5-bisphosphate (PIP₂, blue). (right) Upon mechanical stimulation, the palmitate site is disrupted and the PLD2/TREK-1 complex is released, allowing it to diffuse within the plasma membrane and bind to PIP₂. (bottom) Upon PIP₂ binding, PLD2 is becomes activated and makes phosphatidic acid (PA, red), transducing the mechanical signal into a chemical signal. This chemical signal then binds to and activates TREK-1, initiating chemical-to-electrical signal transduction. (left) When the stimulation is removed, the lipid domain reforms, causing the complex to be re-sequestered. The process results in a return to steady-state conditions, as depicted in the top panel

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Editors

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Reviewer #1 (Public Review):

Force sensing and gating mechanisms of the mechanically activated ion channels is an area of broad interest in the field of mechanotransduction. These channels perform important biological functions by converting mechanical force into electrical signals. To understand their underlying physiological processes, it is important to determine gating mechanisms, especially those mediated by lipids. The authors in this manuscript describe a mechanism for mechanically induced activation of TREK-1 (TWIK-related K⁺ channel). They propose that force induced disruption of ganglioside (GM1) and cholesterol causes relocation of TREK-1 associated with phospholipase D2 (PLD2) to 4,5-bisphosphate (PIP2) clusters, where PLD2 catalytic activity produces phosphatidic acid that can activate the channel. To test their hypothesis, they use dSTORM to measure TREK-1 and PLD2 colocalization with either GM1 or PIP2. They find that shear stress decreases TREK-1/PLD2 colocalization with GM1 and relocates to cluster with PIP2. These movements are affected by TREK-1 C-terminal or PLD2 mutations suggesting that the interaction is important for channel re-location. The authors then draw a correlation to cholesterol suggesting that TREK-1 movement is cholesterol dependent. It is important to note that this is not the only method of channel activation and that one not involving PLD2 also exists. Overall, the authors conclude that force is sensed by ordered lipids and PLD2 associates with TREK-1 to selectively gate the channel.

The proposed mechanism is solid and the authors have revised the manuscript to address previous issues with the first version

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Reviewer #2 (Public Review):

This manuscript by Petersen and colleagues investigates the mechanistic underpinnings of activation of the ion channel TREK-1 by mechanical inputs (fluid shear or membrane stretch) applied to cells. Using a combination of super-resolution microscopy, pair correlation analysis and electrophysiology, the authors convincingly show that the application of shear to a cell can lead to changes in the distribution of TREK-1 and the enzyme PhospholipaseD2 (PLD2), relative to lipid domains defined by either GM1 or PIP2. The activation of TREK-1 by mechanical stimuli was shown to be sensitized by the presence of PLD2, but not a catalytically inactive xPLD2 mutant. In addition, the activity of PLD2 is increased when the molecule is more associated with PIP2, rather than GM1 defined lipid domains. The presented data do not exclude direct mechanical activation of TREK-1, rather suggest a modulation of TREK-1 activity, increasing sensitivity to mechanical inputs, through an inherent mechanosensitivity of PLD2 activity. The authors additionally demonstrate that cellular uptake of cholesterol inhibits TREK-1 activation and, in ex vivo studies, that depletion of cholesterol from astrocytes reduces correlation of TREK-1 and G1 lipids in mouse brain slices. In vivo studies, using *Drosophila melanogaster* behavioural assays, were used to demonstrate that disrupting PLD2 altered behavioural responses to mechanical and electrical inputs.

These data demonstrate that manipulation of PLD2 analogue in the fly can alter sensory transduction, suggesting that PLD functions to regulate sensitivity to mechanical force. However, as the authors note, there is no TREK-1 homologue in this organism: thus the identity of the downstream effectors of PLD in *D. melanogaster* remain unknown. This work will be of interest to the growing community of scientists investigating the myriad mechanisms that can tune mechanical sensitivity of cells, providing valuable insight into the role of functional PLD2 in sensitizing TREK-1 activation in response to mechanical inputs, in some cellular systems.

The authors convincingly demonstrate that, post application of shear, an alteration in the distribution of TREK-1 and mPLD2 (in HEK293T cells) from being correlated with GM1 defined domains (no shear) to increased correlation with PIP2 defined membrane domains (post shear). The association of TREK-1 with PIP2 required functional mPLD2. These data were generated using super-resolution microscopy to visualise, at sub diffraction resolution, the localisation of labelled protein, compared to labelled lipids. The use of super-resolution imaging enabled the authors to visualise changes in cluster association that would not have been achievable with diffraction limited microscopy.

This work provides further evidence of the astounding flexibility of mechanical sensing in cells. By outlining how mechanical activation of TREK-1 can be sensitised by mechanical regulation of PLD2 activity, the authors highlight a mechanism by which TREK-1 sensitivity could be regulated under distinct physiological conditions.

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Reviewer #3 (Public Review):

The manuscript "Mechanical activation of TWIK-related potassium channel by nanoscopic movement and second messenger signaling" presents a new mechanism for the activation of TREK-1 channel. The mechanism suggests that TREK1 is activated by phosphatidic acids that are produced via a mechanosensitive motion of PLD2 to PIP2-enriched domains. Overall, I found the topic interesting but several typos and unclarities reduced the readability of the manuscript. Additionally, I have several major concerns on the interpretation of the results. Therefore, the proposed mechanism is not fully supported by the presented data. Lastly, the mechanism is based on several previous studies from the Hansen lab, however, the novelty of the current manuscript is not clearly stated. For example, in the 2nd result section, the authors stated, "fluid shear causes PLD2 to move from cholesterol dependent GM1 clusters to PIP2 clusters and this activated the enzyme". However, this is also presented as a new finding in section 3 "Mechanism of PLD2 activation by shear."

In the revised manuscript, the authors addressed most of my concerns. I still have the following suggestions/confusions.

1. the reviewer would highly appreciate verification of the cholesterol assay, either by additional experiment or by citations of independent work.
2. The claim on "shear thinning" is still very confusing. First, asymmetric insertion of molecules to one monolayer of the membrane is a main mechanism for membrane bending and curvature formation. Second, why is "shear thinning" equivalent to entropy/order?

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Author Response

The following is the authors' response to the original reviews.

We sincerely thank the reviewers for their in-depth consideration of our manuscript and their helpful reviews. Their efforts have made the paper much better. We have responded to each point. The previously provided public responses have been updated they are included after the private response for convenience.

Reviewer #1 (Recommendations For The Authors):

1. *In general, the manuscript will benefit from copy editing and proof reading. Some obvious edits;*

- *Page 6 line 140. Do the authors mean Cholera toxin B?*

Response: We corrected this error and went through the entire paper carefully correcting for grammar and increased clarity.

- *Page 8 line 173. Methylbetacyclodextrin is misspelled.*

Response: Yes, corrected.

- *Figure 4c is missing representative traces for electrophysiology data.*
- *Figure 4. Please check labeling ordering in figure legend as it does not match the panels in the figure.*

Thank you for the correction and we apologize for the confusion in figure 4. We uploaded an incomplete figure legend, and the old panel 'e' was not from an experiment that was still in the figure. It was removed and the figure legends are now corrected.

- *Please mention the statistical analysis used in all figure legends.*

Response: Thank you for pointing out this omission, statistics have been added.

- *Although the schematics in each figure helps guide readers, they are very inconsistent and sometimes confusing. For example, in Figure 5 the gating model is far-reaching without conclusive evidence, whereas in Figure 6 it is over simplified and unclear what the image is truly representing (granted that the downstream signaling mechanism and channel is not known).*

Response: Figure 5d is the summary figure for the entire paper. We have made this clearer in the figure legend and we deleted the title above the figure that gave the appearance that the panel relates to swell only. It is the proposed model based on what we show in the paper and what is known about the activation mechanism of TREK-1.

Figure 6 is supposed to be simple. It is to help the reader understand that when PA is low mechanical sensitivity is high. Without the graphic, previous reviewers got confused about threshold going down and mechanosensitivity going up and how the levels of PA relate. Low PA= high sensitivity. We've added a downstream effector to the right side of the panel to avoid any biased to a putative downstream channel effector. The purpose of the experiment is to show PLD has a mechanosensitive phenotype in vivo.

Reviewer #2 (Recommendations For The Authors):

This manuscript outlines some really interesting findings demonstrating a mechanism by which mechanically driven alterations in molecular distributions can influence a) the activity of the PLD2 molecule and subsequently b) the activation of TREK-1 when mechanical inputs are applied to a cell or cell membrane.

The results presented here suggest that this redistribution of molecules represents a modulatory mechanism that alters either the amplitude or the sensitivity of TREK-1 mediated currents evoked by membrane stretch. While the authors do present values for the pressure required to activate 50% of channels (P50), the data presented provides incomplete evidence to conclude a shift in threshold of the currents, given that many of the current traces provided in the supplemental material do not saturate within the stimulus range, thus limiting the application of a Boltzmann fit to determine the P50. I suggest adding additional context to enable readers to better assess the limitations of this use of the Boltzmann fit to generate a P50, or alternately repeating the experiments to apply stimuli up to lytic pressures to saturate the mechanically evoked currents, enabling use of the Boltzmann function to fit the data.

Response: We thank the reviewer for pointing this out. We agree the currents did not reach saturation. Hence the term P50 could be misleading, so we have removed it from the paper. We now say “half maximal” current measured from non-saturating pressures of 0-60 mmHg. We also deleted the xPLD data in supplemental figure 3C since there is insufficient current to realistically estimate a half maximal response.

In my opinion, the conclusions presented in this manuscript would be strengthened by an assessment of the amount of TREK-1 in the plasma membrane pre and post application of shear. While the authors do present imaging data in the supplementary materials, these data are insufficiently precise to comment on expression levels in the membrane. To strengthen this conclusion the authors could conduct cell surface biotinylation assays, as a more sensitive and quantitative measure of membrane localisation of the proteins of interest.

1. Response: as mentioned previously, we do not have an antibody to the extracellular domain. Nonetheless to better address this concern we directly compared the levels of TREK-1, PIP2, and GM1; in xPLD2, mPLD2, enPLD2 with and without shear. The results are in supplemental figure 2. PLD2 is known to increase endocytosis¹ and xPLD2 is known to block both agonist induced and constitutive endocytosis of μ -opioid receptor². The receptor is trapped on the surface. This is true of many proteins including Rho3, ARF4, and ACE21 among others. In agreement with this mechanism, in Figure S2C,G we show that TREK increases with xPLD and the localization can clearly be seen at the plasma membrane just like in all of the other publications with xPLD overexpression. xPLD2 would be expected to inhibit the basal current but we presume the increased expression likely has compensated and there is sufficient PA and PG from other sources to allow for the basal current. It is in this state that we then conduct our ephys and monitor with a millisecond time resolution and see no activation. We are deriving conclusion from a very clear response—Figure 1b shows almost no current, even at 1-10 ms after applying pressure. There is little pressure current when we know the channel is present and capable of conducting ion (Figure 1d red bar). After shear there is a strong decrease in TREK-1 currents on the membrane in the presence of xPLD2. But it is not less than TREK-1 expression with mPLD2. And since mouse PLD2 has the highest basal current and pressure activation current. The amount of TREK-1 present is sufficient to conduct large current. To have almost no detectable current would require at least a 10 fold reduction compared to mPLD2 levels before we would lack the sensitivity to see a channel open. Lastly endocytosis typically in on the order of seconds to minutes, no milliseconds.
2. We have shown an addition 2 independent ways that TREK-1 is on the membrane during our stretch experiments. Figure 1d shows the current immediately prior to applying pressure for wt TREK-1. When catalytically dead PLD is present (xPLD2) there is almost normal basal current. The channel is clearly present. And then in figure 1a we show within a millisecond there is no pressure current. As a control we added a functionally dead TREK-1 truncation (xTREK). Compared to xPLD2 there is clearly normal basal current. If this is not strong evidence the channel was available on the surface for mechanical activation please help us understand why. And if you think within 2.1 ms 100% of the channel is gone by endocytosis please provide some evidence that this is possible so we can reconsider.
3. We have TIRF super resolution imaging with ~20 nm x-y resolution and ~ 100nm z resolution and Figure 2b clearly shows the channel on the membrane. When we apply pressure in 1b, the channel is present.
4. Lastly, In our previous studies we showed activation of PLD2 by anesthetics was responsible for all of TREK-1's anesthetic sensitivity and this was through PLD2 binding to the C-terminus of TREK-15. We showed this was the case by transferring anesthetic sensitivity to an anesthetic insensitive homolog TRAAK. This established conclusively the basic premise of our mechanism. Here we show the same C-terminal region and PLD2 are responsible for the mechanical current observed by TREK-1. TRAAK is already mechanosensitive so the same chimera will not work for our purposes here. But anesthetic activation and mechanical activation are dramatically different stimuli, and the fact that the role of PLD is robustly observed in both should be considered.

The authors discuss that the endogenous levels of TREK-1 and PLD2 are "well correlated: in C2C12 cells, that TREK-1 displayed little pair correlation with GM1 and that a "small amount of TREK-1 trafficked to PIP2". As such, these data suggest that the data outlined for HEK293T cells may be hampered by artefacts arising from overexpression. Can TREK-

1 currents be activated by membrane stretch in these cells C2C12 cells and are they negatively impacted by the presence of xPLD2? Answering this question would provide more insight into the proposed mechanism of action of PLD2 outlined by the authors in this manuscript. If no differences are noted, the model would be called into question. It could be that there are additional cell-specific factors that further regulate this process.

Response: The low pair correlation of TREK-1 and GM1 in C2C12 cells was due to insufficient levels of cholesterol in the cell membrane to allow for robust domain formation. In Figure 4b we loaded C2C12 cells with cholesterol using the endogenous cholesterol transport protein apoE and serum (an endogenous source of cholesterol). As can be seen in Fig. 4b, the pair correlation dramatically increased (purple line). This was also true in neuronal cells (N2a) (Fig 4d, purple bar). And shear (3 dynes/cm²) caused the TREK-1 that was in the GM1 domains to leave (red bar) reversing the effect of high cholesterol. This demonstrates our proposed mechanism is working as we expect with endogenously expressed proteins.

There are many channels in C2C12 cells, it would be difficult to isolate TREK-1 currents, which is why we replicated the entire system (ephys and dSTORM) in HEK cells. Note, in figure 4c we also show that adding cholesterol inhibits TREK-1 whole cell currents in HEK293cells.

As mentioned in the public review, the behavioural experiments in D. melanogaster can not solely be attributed to a change in threshold. While there may be a change in the threshold to drive a different behaviour, the writing is insufficiently precise to make clear that conclusions cannot be drawn from these experiments regarding the functional underpinnings of this outcome. Are there changes in resting membrane potential in the mutant flies? Alterations in Nav activity? Without controlling for these alternate explanations it is difficult to see what this last piece of data adds to the manuscript, particularly given the lack of TREK-1 in this organism. At the very least, some editing of the text to more clearly indicate that these data can only be used to draw conclusions on the change in threshold for driving the behaviour not the change in threshold of the actual mechanotransduction event (i.e. conversion of the mechanical stimulus into an electrochemical signal).

Response: We agree; features other than PLDs direct mechanosensitivity are likely contributing. This was shown in figure 6g left side. We have an arrow going to ion channel and to other downstream effectors. We've added the putative alteration to downstream effectors to the right side of the panel. This should make it clear that we no more speculate the involvement of a channel than any of the other many potential downstream effectors. As mentioned above, the figure helps the reader coordinate low PA with increased mechanosensitivity. Without the graphic reviewers got confused that PA increased the threshold which corresponds to a decreased sensitivity to pain. Nonetheless we removed our conclusion about fly thresholds from the abstract and made clearer in the main text the lack of mechanism downstream of PLD in flies including endocytosis. Supplemental Figure S2H also helps emphasize this. .

Nav channels are interesting, and since PLD contribute to endocytosis and Nav channels are also regulated by endocytosis there is likely a PLD specific effect using Nav channels. There are many ways PA likely regulates mechanosensitive thresholds, but we feel Nav is beyond the scope of our paper. Someone else will need to do those studies. We have amended a paragraph in the conclusion which clearly states we do not know the specific mechanism at work here with the suggestions for future research to discover the role of lipid and lipid-modifying enzymes in mechanosensitive neurons.

There may be fundamental flaws in how the statistics have been conducted. The methods section indicates that all statistical testing was performed with a Student's t-test. A visual

scan of many of the data sets in the figures suggests that they are not normally distributed, thus a parametric test such as a Student's t-test is not valid. The authors should assess if each data set is normally distributed, and if not, a non-parametric statistical test should be applied. I recommend assessing the robustness of the statistical analyses and adjusting as necessary.

Response: We thank the reviewer for pointing this out, indeed there is some asymmetry in Figure 6C-d. The p values with Mann Whitney were slightly improved $p=0.016$ and $p=0.0022$ for 6c and 6d respectively. For reference, the students t-test had slightly worse statistics $p=0.040$ and $p=0.0023$. The score remained the same 1 and 2 stars respectively.

The references provided for the statement regarding cascade activation of the TRPs are incredibly out of date. While it is clear that TRPV4 can be activated by a second messenger cascade downstream of osmotic swelling of cells, TRPV4 has also been shown to be activated by mechanical inputs at the cell-substrate interface, even when the second messenger cascade is inhibited. Recommend updating the references to reflect more current understanding of channel activation.

Response: We thank the reviewer for pointing this out. We have updated the references and changed the comment to “can be” instead of “are”. The reference is more general to multiple ion channel types including KCNQ4. This should avoid any perceived conflict with the cellsubstrate interface mechanism which we very much agree is a correct mechanism for TRP channels.

Minor comments re text editing etc:

The central messages of the manuscript would benefit from extensive work to increase the precision of the writing of the manuscript and the presentation of data in the figures, such textual changes alone would help address a number of the concerns outlined in this review, by clarifying some ambiguities. There are numerous errors throughout, ranging from grammatical issues, ambiguities with definitions, lack of scale bars in images, lack of labels on graph axes, lack of clarity due to the mode of presentation of sample numbers (it would be far more precise to indicate specific numbers for each sample rather than a range, which is ambiguous and confusing), unnecessary and repeat information in the methods section. Below are some examples but this list is not exhaustive.

Response: Thank you, reviewer # 1 also had many of these concerns. We have gone through the entire paper and improved the precision of the writing of the manuscript. We have also added the missing error bar to Figure 6. And axis labels have been added to the inset images. The redundancy in cell culture methods has been removed. Where a range is small and there are lots of values, the exact number of ‘n’ are graphically displayed in the dot plot for each condition.

Text:

I recommend considering how to discuss the various aspects of channel activation. A convention in the field is to use mechanical activation or mechanical gating to describe that process where the mechanical stimulus is directly coupled to the channel gating mechanism. This would be the case for the activation of TREK-1 by membrane stretch alone. The increase in activation by PLD2 activity then reflects a modulation of the mechanical activation of the channel, because the relevant gating stimulus is PA, rather than force/stretch. The sum of these events could be described as shear-evoked or mechanically-evoked, TREK-1 mediated currents (thus making it clear that the mechanical stimulus initiates the relevant cascade, but the gating stimulus may be other

than direct mechanical input.) Given the interesting and compelling data offered in this manuscript regarding the sensitisation of TREK-1 dependent mechanically evoked currents by PLD2, an increase in the precision of the language would help convey the central message of this work.

Response; We agree there needs to be convention. We have taken the suggestion of mechanically evoked and we suggest the following definitions:

1. Mechanical activation of PLD2: direct force on the lipids releasing PLD2 from nonactivating lipids.
2. Mechanical activation/gating of TREK1: direct force from lipids from either tension or hydrophobic mismatch that opens the channel.
3. Mechanically evoked: a mechanical event that leads to a downstream effect. The effect is mechanically “evoked”.
4. Spatial patterning/biochemistry: nanoscopic changes in the association of a protein with a nanoscopic lipid cluster or compartment.

An example of where discussion of mechanical activation is ambiguous in the text is found at line 109: "channel could be mechanically activated by a movement from GM1 to PIP2 lipids." In this case, the sentence could be suggesting that the movement between lipids provides the mechanical input that activates the channel, which is not what the data suggest.

Response: Were possible we have replaced “movement” with “spatial patterning” and “association” and “dissociation” from specific lipid compartment. This better reflects the data we have in this paper. However, we do think that a movement mechanically activates the channel, GM1 lipids are thick and PIP2 lipids are thin, so movement between the lipids could activate the channel through direct lipid interaction. We will address this aspect in a future paper.

Inconsistencies with usage:

- *TREK1 versus TREK-1*

Response: corrected to TREK-1

- *mPLD2 versus PLD2*

Response: where PLD2 represents mouse this has been corrected.

- *K758R versus xPLD2*

Response: we replaced K758R in the methods with xPLD2.

- *HEK293T versus HEK293t Response: we have changed all instances to read HEK293T.*
- *Drosophila melanogaster and D. melanogaster used inconsistently and in many places incorrectly*

Response: we have read all to read the common name Drosophila.

Line 173: misspelled methylbetacyclodextrin

Response corrected

| *Line 174: degree symbol missing*

Response corrected

| *Line 287: "the decrease in cholesterol likely evolved to further decrease the palmitate order in the palmitate binding site"... no evidence, no support for this statement, falsely attributes intention to evolutionary processes .*

Response: we have removed the reference to evolution at the request of the reviewer, it is not necessary. But we do wish to note that to our knowledge, all biological function is scientifically attributed to evolution. The fact that cholesterol decreases in response to shear is evidence alone that the cell evolved to do it.

| *Line 307: grammatical error*

Response: the redundant Lipid removed.

| *Line 319: overinterpreted - how is the mechanosensitivity of GPCRs explained by this translocation?*

Response: all G-alpha subunits of the GPCR complex are palmitoylated. We showed PLD (which has the same lipidation) is mechanically activated. If the palmitate site is disrupted for PLD2, then it is likely disrupted for every G-alpha subunit as well.

| *Line 582: what is the wild type referred to here?*

Response: human full length with a GFP tag.

| *Methods:*

| *• Sincere apologies if I missed something but I do not recall seeing any experiments using purified TREK-1 or flux assays. These details should be removed from the methods section*

Response: Removed.

| *• There is significant duplication of detail across the methods (three separate instances of electrophysiology details) these could definitely be consolidated.*

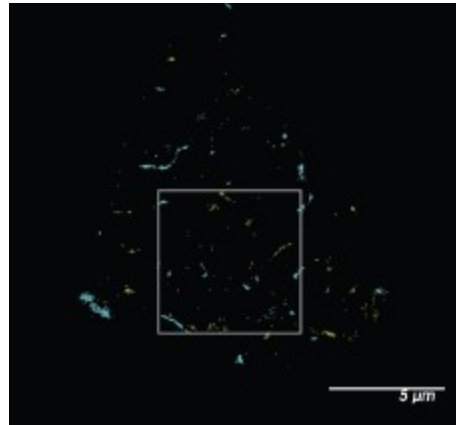
Response: Duplicates removed.

| *Figures:*

| *• Figure 2- b box doesn't correspond to inset. Bottom panel should provide overview image for the cell that was assessed with shear. In bottom panel, circle outlines an empty space.*

Response: We have widened the box slightly to correspond so the non shear box corresponds to the middle panel. We have also added the picture for the whole cell to Fig S2g and outlined the zoom shown in the bottom panel of Fig 2b as requested. The figure is of the top of a cell. We also added the whole cell image of a second sheared cell.

Author response image 1.



- *Figure 3 b+c: inset graph lacking axis labels*

Response; the inset y axis is the same as the main axis. We added “pair corr. (5nM)” and a description in the figure legend to make this clearer. The purpose of the inset is to show statistical significance at a single point. The contrast has been maximized but without zooming in points can be difficult to see.

- *Figure 5: replicate numbers missing and individual data points lacking in panels b + c, no labels of curve in b + c, insets, unclear what (5 nm) refers to in insets.*

Response: Thank you for pointing out these errors. The N values have been added. Similar to figure 3, the inset is a bar graph of the pair correlation data at 5 nm. A better explanation of the data has been added to the figure legend.

- *Figure 6: no scale bar, no clear membrane localization evident from images presented, panel g offers virtually nothing in terms of insight*

Response: We have added scale bars to figure 6b. Figure 6g is intentionally simplistic, we found that correlating decreased threshold with increased pain was confusing. A previous reviewer claimed our data was inconsistent. The graphic avoids this confusion. We also added negative effects of low PA on downstream effects to the right panel. This helps graphically show we don't know the downstream effects.

Reviewer #3 (Recommendations For The Authors):

Minor suggestions:

1. line 162, change 'heat' to 'temperature'.

Response: changed.

1. in figure 1, it would be helpful to keep the unit for current density consistent among different panels. 1e is a bit confusing: isn't the point of Figure 1 that most of TREK1 activation is not caused by direct force-sensing?

Response: Yes, the point of figure 1 is to show that in a biological membrane over expressed TREK-1 is a downstream effector of PLD2 mechanosensation which is indirect. We agree the figure legend in the previous version of the paper is very confusing.

There is almost no PLD2 independent current in our over expressed system, which is represented by no ions in the conduction pathway of the channel despite there being tension on the membrane.

Purified TREK-1 is only mechanosensitive in a few select lipids, primarily crude Soy PC. It was always assumed that HEK293 and Cos cells had the correct lipids since over expressed TREK-1 responded to mechanical force in these lipids. But that does not appear to be correct, or at least only a small amount of TREK-1 is in the mechanosensitive lipids. Figure 1e graphically shows this. The arrows indicate tension, but the channel isn't open with xPLD2 present. We added a few sentences to the discussion to further clarify.

Panels c has different units because the area of the tip was measured whereas in d the resistance of the tip was measured. They are different ways for normalizing for small differences in tip size.

| 1. line 178, ~45 of what?

Response: Cells were fixed for ~30 sec.

| 1. line 219 should be Figure 4f?

Response: thank you, yes Figure 4f.

Previous public reviews with minor updates.

Reviewer #1 (Public Review):

Force sensing and gating mechanisms of the mechanically activated ion channels is an area of broad interest in the field of mechanotransduction. These channels perform important biological functions by converting mechanical force into electrical signals. To understand their underlying physiological processes, it is important to determine gating mechanisms, especially those mediated by lipids. The authors in this manuscript describe a mechanism for mechanically induced activation of TREK-1 (TWIK-related K⁺ channel. They propose that force induced disruption of ganglioside (GM1) and cholesterol causes relocation of TREK-1 associated with phospholipase D2 (PLD2) to 4,5-bisphosphate (PIP2) clusters, where PLD2 catalytic activity produces phosphatidic acid that can activate the channel. To test their hypothesis, they use dSTORM to measure TREK-1 and PLD2 colocalization with either GM1 or PIP2. They find that shear stress decreases TREK-1/PLD2 colocalization with GM1 and relocates to cluster with PIP2. These movements are affected by TREK-1 C-terminal or PLD2 mutations suggesting that the interaction is important for channel re-location. The authors then draw a correlation to cholesterol suggesting that TREK-1 movement is cholesterol dependent. It is important to note that this is not the only method of channel activation and that one not involving PLD2 also exists. Overall, the authors conclude that force is sensed by ordered lipids and PLD2 associates with TREK-1 to selectively gate the channel. Although the proposed mechanism is solid, some concerns remain.

1. Most conclusions in the paper heavily depend on the dSTORM data. But the images provided lack resolution. This makes it difficult for the readers to assess the representative images.

Response: The images were provided are at 300 dpi. Perhaps the reviewer is referring to contrast in Figure 2? We are happy to increase the contrast or resolution.

As a side note, we feel the main conclusion of the paper, mechanical activation of TREK-1 through PLD2, depended primarily on the electrophysiology in Figure 1b-c, not the dSTORM. But both complement each other.

1. The experiments in Figure 6 are a bit puzzling. The entire premise of the paper is to establish gating mechanism of TREK-1 mediated by PLD2; however, the motivation behind using flies, which do not express TREK-1 is puzzling.

Response: The fly experiment shows that PLD mechanosensitivity is more evolutionarily conserved than TREK-1 mechanosensitivity. We have added this observation to the paper.

-Figure 6B, the image is too blown out and looks over saturated. Unclear whether the resolution in subcellular localization is obvious or not.

Response: Figure 6B is a confocal image, it is not dSTORM. There is no dSTORM in Figure 6. We have added the error bars to make this more obvious. For reference, only a few cells would fit in the field of view with dSTORM.

-Figure 6C-D, the differences in activity threshold is 1 or less than 1g. Is this physiologically relevant? How does this compare to other conditions in flies that can affect mechanosensitivity, for example?

Response: Yes, 1g is physiologically relevant. It is almost the force needed to wake a fly from sleep (1.2-3.2g). See ref 33. Murphy Nature Pro. 2017.

1. 70mOsm is a high degree of osmotic stress. How confident are the authors that a cell health is maintained under this condition and b. this does indeed induce membrane stretch? For example, does this stimulation activate TREK-1?

Response: Yes, osmotic swell activates TREK1. This was shown in ref 19 (Patel et al 1998). We agree the 70 mOsm is a high degree of stress. This needs to be stated better in the paper.

Reviewer #2 (Public Review):

This manuscript by Petersen and colleagues investigates the mechanistic underpinnings of activation of the ion channel TREK-1 by mechanical inputs (fluid shear or membrane stretch) applied to cells. Using a combination of super-resolution microscopy, pair correlation analysis and electrophysiology, the authors show that the application of shear to a cell can lead to changes in the distribution of TREK-1 and the enzyme PhospholipaseD2 (PLD2), relative to lipid domains defined by either GM1 or PIP2. The activation of TREK-1 by mechanical stimuli was shown to be sensitized by the presence of PLD2, but not a catalytically dead xPLD2 mutant. In addition, the activity of PLD2 is increased when the molecule is more associated with PIP2, rather than GM1 defined lipid domains. The presented data do not exclude direct mechanical activation of TREK-1, rather suggest a modulation of TREK-1 activity, increasing sensitivity to mechanical inputs, through an inherent mechanosensitivity of PLD2 activity. The authors additionally claim that PLD2 can regulate transduction thresholds in vivo using Drosophila melanogaster behavioural assays. However, this section of the manuscript overstates the experimental findings, given that it is unclear how the disruption of PLD2 is leading to behavioural changes, given the lack of a TREK-1 homologue in this organism and the lack of supporting data on molecular function in the relevant cells.

Response: We agree, the downstream effectors of PLD2 mechanosensitivity are not known in the fly. Other anionic lipids have been shown to mediate pain see ref 46 and 47. We do not wish to make any claim beyond PLD2 being an in vivo contributor to a fly's response to mechanical force. We have removed the speculative conclusions about fly thresholds from the abstract.

That said we do believe we have established a molecular function at the cellular level. We showed PLD is robustly mechanically activated in a cultured fly cell line (BG2-c2) Figure 6a of the manuscript. And our previous publication established mechanosensation of PLD (Petersen et. al. Nature Com 2016) through mechanical disruption of the lipids. At a minimum, the experiments show PLDs mechanosensitivity is evolutionarily better conserved across species than TREK1.

This work will be of interest to the growing community of scientists investigating the myriad mechanisms that can tune mechanical sensitivity of cells, providing valuable insight into the role of functional PLD2 in sensi>zing TREK-1 activation in response to mechanical inputs, in some cellular systems.

The authors convincingly demonstrate that, post application of shear, an alteration in the distribution of TREK-1 and mPLD2 (in HEK293T cells) from being correlated with GM1 defined domains (no shear) to increased correlation with PIP2 defined membrane domains (post shear). These data were generated using super-resolution microcopy to visualise, at sub diffraction resolution, the localisation of labelled protein, compared to labelled lipids. The use of super-resolution imaging enabled the authors to visualise changes in cluster association that would not have been achievable with diffraction limited microcopy. However, the conclusion that this change in association reflects TREK-1 leaving one cluster and moving to another overinterprets these data, as the data were generated from sta>c measurements of fixed cells, rather than dynamic measurements capturing molecular movements.

When assessing molecular distribution of endogenous TREK-1 and PLD2, these molecules are described as "well correlated: in C2C12 cells" however it is challenging to assess what "well correlated" means, precisely in this context. This limitation is compounded by the conclusion that TREK-1 displayed little pair correlation with GM1 and the authors describe a "small amount of TREK-1 trafficked to PIP2". As such, these data may suggest that the findings outlined for HEK293T cells may be influenced by artefacts arising from overexpression.

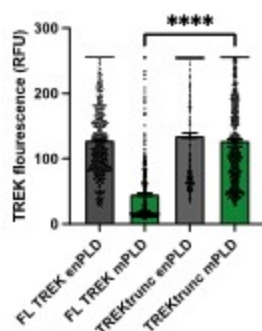
The changes in TREK-1 sensitivity to mechanical activation could also reflect changes in the amount of TREK-1 in the plasma membrane. The authors suggest that the presence of a leak currently accounts for the presence of TREK-1 in the plasma membrane, however they do not account for whether there are significant changes in the membrane localisation of the channel in the presence of mPLD2 versus xPLD2. The supplementary data provide some images of fluorescently labelled TREK-1 in cells, and the authors state that truncating the c-terminus has no effect on expression at the plasma membrane, however these data provide inadequate support for this conclusion. In addition, the data reporting the P50 should be noted with caution, given the lack of saturation of the current in response to the stimulus range.

Response: We thank the reviewer for his/her concern about expression levels. We did test TREK-1 expression. mPLD decreases TREK-1 expression ~two-fold (see Author response image 2 below). We did not include the mPLD data since TREK-1 was mechanically activated with mPLD. For expression to account for the loss of TREK-1 stretch current (Figure 1b), xPLD would need to block surface expression of TREK-1 prior to stretch. The opposite was true, xPLD2 increased TREK-1 expression (see Figure S2c). Furthermore, we tested the leak current

of TREK-1 at 0 mV and 0 mmHg of stretch. Basal leak current was no different with xPLD2 compared to endogenous PLD (Figure 1d; red vs grey bars respectively) suggesting TREK-1 is in the membrane and active when xPLD2 is present. If anything, the magnitude of the effect with xPLD would be larger if the expression levels were equal.

Author response image 2.

TREK expression at the plasma membrane. TREK-1 Fluorescence was measured by GFP at points along the plasma membrane. Over expression of mouse PLD2 (mPLD) decrease the amount of full-length TREK-1 (FL TREK) on the surface more than 2-fold compared to endogenously expressed PLD (enPLD) or truncated TREK (TREKtrunc) which is missing the PLD binding site in the C-terminus. Over expression of mPLD had no effect on TREKtrunc.



Finally, by manipulating PLD2 in D. melanogaster, the authors show changes in behaviour when larvae are exposed to either mechanical or electrical inputs. The depletion of PLD2 is concluded to lead to a reduction in activation thresholds and to suggest an in vivo role for PA lipid signaling in setting thresholds for both mechanosensitivity and pain. However, while the data provided demonstrate convincing changes in behaviour and these changes could be explained by changes in transduction thresholds, these data only provide weak support for this specific conclusion. As the authors note, there is no TREK-1 in D. melanogaster, as such the reported findings could be accounted for by other explanations, not least including potential alterations in the activation threshold of Nav channels required for action potential generation. To conclude that the outcomes were in fact mediated by changes in mechanotransduction, the authors would need to demonstrate changes in receptor potential generation, rather than deriving conclusions from changes in behaviour that could arise from alterations in resting membrane potential, receptor potential generation or the activity of the voltage gated channels required for action potential generation.

Response: We are willing to restrict the conclusion about the fly behavior as the reviewers see fit. We have shown PLD is mechanosensitivity in a fly cell line, and when we knock out PLD from a fly, the animal exhibits a mechanosensation phenotype. We tried to make it clear in the figure and in the text that we have no evidence of a particular mechanism downstream of PLD mechanosensation.

This work provides further evidence of the astounding flexibility of mechanical sensing in cells. By outlining how mechanical activation of TREK-1 can be sensitised by mechanical regulation of PLD2 activity, the authors highlight a mechanism by which TREK-1 sensitivity could be regulated under distinct physiological conditions.

Reviewer #3 (Public Review):

The manuscript "Mechanical activation of TWIK-related potassium channel by nanoscopic movement and second messenger signaling" presents a new mechanism for the activation of TREK-1 channel. The mechanism suggests that TREK1 is activated by phosphatidic acids that are produced via a mechanosensitive motion of PLD2 to PIP2-enriched domains. Overall, I found the topic interesting, but several typos and unclarities reduced the readability of the manuscript. Additionally, I have several major concerns on the interpretation of the results. Therefore, the proposed mechanism is not fully supported by the presented data. Lastly, the mechanism is based on several previous studies from the Hansen lab, however, the novelty of the current manuscript is not clearly stated. For example, in the 2nd result section, the authors stated, "fluid shear causes PLD2 to move from cholesterol dependent GM1 clusters to PIP2 clusters and this activated the enzyme". However, this is also presented as a new finding in section 3 "Mechanism of PLD2 activation by shear."

For PLD2 dependent TREK-1 activation. Overall, I found the results compelling. However, two key results are missing.

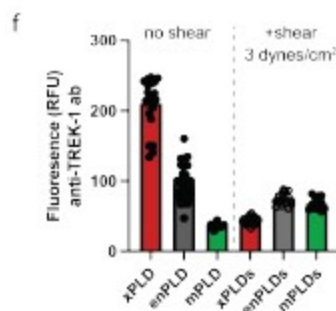
- 1. Does HEK cells have endogenous PLD2? If so, it's hard to claim that the authors can measure PLD2-independent TREK1 activation.*

Response: yes, there is endogenous PLD (enPLD). We calculated the relative expression of xPLD2 vs enPLD. xPLD2 is >10x more abundant (Fig. S3d of Pavel et al PNAS 2020, ref 14 of the current manuscript). Hence, as with anesthetic sensitivity, we expect the xPLD to out compete the endogenous PLD, which is what we see. We added the following sentence and reference : "The xPLD2 expression is >10x the endogenous PLD2 (enPLD2) and out competes the TREK-1 binding site for PLD25."

- 1. Does the plasma membrane trafficking of TREK1 remain the same under different conditions (PLD2 overexpression, truncation)? From Figure S2, the truncated TREK1 seem to have very poor trafficking. The change of trafficking could significantly contribute to the interpretation of the data in Figure 1.*

Response: If the PLD2 binding site is removed (TREK-1trunc), yes, the trafficking to the plasma membrane is unaffected by the expression of xPLD and mPLD (Author response image 2 above). For full length TREK1 (FL-TREK-1), co-expression of mPLD decreases TREK expression (Author response image 2) and coexpression with xPLD increases TREK expression (Figure S2f). This is exactly opposite of what one would expect if surface expression accounted for the change in pressure currents. Hence, we conclude surface expression does not account for loss of TREK-1 mechanosensitivity with xPLD2. A few sentences was added to the discussion. We also performed dSTORM on the TREKtruncated using EGFP. TREK-truncated goes to PIP2 (see figure 2 of 6)

Author response image 3.



To better compare the levels of TREK-1 before and after shear, we added a supplemental figure S2f where the protein was compared simultaneously in all conditions. 15 min of shear significantly decreased TREK-1 except with mPLD2 where the levels before shear were already lowest of all the expression levels tested.

For shear-induced movement of TREK1 between nanodomains. The section is convincing, however I'm not an expert on super-resolution imaging. Also, it would be helpful to clarify whether the shear stress was maintained during fixation. If not, what is the >me gap between reduced shear and the fixed state. lastly, it's unclear why shear flow changes the level of TREK1 and PIP2.

Response: Shear was maintained during the fixing. xPLD2 blocks endocytosis, presumably endocytosis and or release of other lipid modifying enzymes affect the system. The change in TREK-1 levels appears to be directly through an interaction with PLD as TREK trunc is not affected by over expression of xPLD or mPLD.

For the mechanism of PLD2 activation by shear. I found this section not convincing. Therefore, the question of how does PLD2 sense mechanical force on the membrane is not fully addressed. Par>cularly, it's hard to imagine an acute 25% decrease cholesterol level by shear - where did the cholesterol go? Details on the measurements of free cholesterol level is unclear and additional/alternative experiments are needed to prove the reduction in cholesterol by shear.

Response: The question “how does PLD2 sense mechanical force on the membrane” we addressed and published in Nature Comm. In 2016. The title of that paper is “Kinetic disruption of lipid rafts is a mechanosensor for phospholipase D” see ref 13 Petersen et. al. PLD is a soluble protein associated to the membrane through palmitoylation. There is no transmembrane domain, which narrows the possible mechanism of its mechanosensation to disruption.

The Nature Comm. reviewer identified as “an expert in PLD signaling” wrote the following of our data and the proposed mechanism:

“This is a provocative report that identi0ies several unique properties of phospholipase D2 (PLD2). It explains in a novel way some long established observations including that the enzyme is largely regulated by substrate presentation which 0its nicely with the authors model of segregation of the two lipid raft domains (cholesterol ordered vs PIP2 containing). Although PLD has previously been reported to be involved in mechanosensory transduction processes (as cited by the authors) this is the 0irst such report associating the enzyme with this type of signaling... It presents a novel model that is internally consistent with previous

literature as well as the data shown in this manuscript. It suggests a new role for PLD2 as a force transduction tied to the physical structure of lipid rafts and uses parallel methods of disruption to test the predictions of their model.”

Regarding cholesterol. We use a fluorescent cholesterol oxidase assay which we described in the methods. This is an appropriate assay for determining cholesterol levels in a cell which we use routinely. We have published in multiple journals using this method, see references 28, 30, 31. Working out the metabolic fate of cholesterol after shear is indeed interesting but well beyond the scope of this paper. Furthermore, we indirectly confirmed our finding using dSTORM cluster analysis (Figure 3d-e). The cluster analysis shows a decrease in GM1 cluster size consistent with our previous experiments where we chemically depleted cholesterol and saw a similar decrease in cluster size (see ref 13). All the data are internally consistent, and the cholesterol assay is properly done. We see no reason to reject the data.

Importantly, there is no direct evidence for "shear thinning" of the membrane and the authors should avoid claiming shear thinning in the abstract and summary of the manuscript.

Response: We previously established a kinetic model for PLD2 activation see ref 13 (Petersen et al Nature Comm 2016). In that publication we discussed both entropy and heat as mechanisms of disruption. Here we controlled for heat which narrowed that model to entropy (i.e., shear thinning) (see Figure 3c). We provide an overall justification below. But this is a small refinement of our previous paper, and we prefer not to complicate the current paper. We believe the proper rheological term is shear thinning. The following justification, which is largely adapted from ref 13, could be added to the supplement if the reviewer wishes.

Justification: To establish shear thinning in a biological membrane, we initially used a soluble enzyme that has no transmembrane domain, phospholipase D2 (PLD2). PLD2 is a soluble enzyme and associated with the membrane by palmitate, a saturated 16 carbon lipid attached to the enzyme. In the absence of a transmembrane domain, mechanisms of mechanosensation involving hydrophobic mismatch, tension, midplane bending, and curvature can largely be excluded. Rather the mechanism appears to be a change in fluidity (i.e., kinetic in nature). GM1 domains are ordered, and the palmitate forms van der Waals bonds with the GM1 lipids. The bonds must be broken for PLD to no longer associate with GM1 lipids. We established this in our 2016 paper, ref 13. In that paper we called it a kinetic effect, however we did not experimentally distinguish enthalpy (heat) vs. entropy (order). Heat is Newtonian and entropy (i.e., shear thinning) is non-Newtonian. In the current study we paid closer attention to the heat and ruled it out (see Figure 3c and methods). We could propose a mechanism based on kinetic disruption, but we know the disruption is not due to melting of the lipids (enthalpy), which leaves shear thinning (entropy) as the plausible mechanism.

The authors should also be aware that hypotonic shock is a very dirty assay for stretching the cell membrane. Ouen, there is only a transient increase in membrane tension, accompanied by many biochemical changes in the cells (including acidification, changes of concentration etc). Therefore, I would not consider this as definitive proof that PLD2 can be activated by stretching membrane.

Response: Comment noted. We trust the reviewer is correct. In 1998 osmotic shock was used to activate the channel. We only intended to show that the system is consistent with previous electrophysiologic experiments.

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